Semiclassical Gravitation of Prefractal Casimir Cavities: Macroscopic Effective Traces, Vernier Gravitational Radiation, Energy-Condition Switching, and Vacuum Frame Dragging

Goren Gordon*
Department of Informatics, Indiana University Bloomington, Bloomington, Indiana 47408, USA
(Dated: November 13, 2026)

Between infinite, flat, ideally conducting parallel plates in $(3+1)$-dimensional Minkowski spacetime, the renormalized electromagnetic Casimir stress-energy tensor $\langle\hat{T}_{\mu\nu}\rangle_{\rm ren}$ is conformally traceless ($T^{\mu}{}_{\mu}=0$) and invariant under in-plane Lorentz boosts ($\rho_{\rm vac}c^2+P_{\parallel}=0$). Coupled to semiclassical Einstein general relativity (GR), $G_{\mu\nu}=(8\pi G/c^4)\langle\hat{T}_{\mu\nu}\rangle_{\rm ren}$, the canonical planar vacuum therefore sources zero local Ricci scalar curvature ($R=0$), decouples from scalar-tensor breathing modes in bulk vacuum, and carries zero non-dissipative angular momentum under rigid in-plane rotation. Here we formulate the active semiclassical gravitational backreaction of an idealized zero-temperature *Canonical Prefractal Sector Cavity* (CPSC)—an azimuthally sectored, radially self-similar boundary hierarchy of finite generation N with short-distance length cutoff $\ell_*=\Delta r\, b^{-N}$ that endows the Casimir coefficient with discrete scale dependence (log-periodically running with scale ratio b) while breaking planar boost symmetry and modulating boundary conditions between open-aperture saddle-point fringing and parallel-sector confinement. Rather than equating a postulated cavity-integrated trace to pointwise Ricci curvature, we reconstruct the local quantum field theory (QFT) expectation value $\langle\hat{T}_{\mu\nu}(\mathbf{x})\rangle_{\rm ren}$ from first-principles cyclic-Bloch three-dimensional boundary-element method (BEM) scattering across conformally coupled scalar ($\xi=1/6$), minimally coupled scalar ($\xi=0$), and perfect electric conductor (PEC) vector electromagnetic fields, and solve the full linearized Einstein, Starobinsky $f(R)=R+\alpha R^2$ ($\alpha>0$), and gravito-electromagnetic (GEM) field equations in Planck-normalized dimensionless units ($\epsilon_P=\ell_P/L$). Enforcing exact covariant conservation $\nabla_\mu(T^{\mu\nu}_{\rm vac}+T^{\mu\nu}_{\rm con})=0$ in the sub-relativistic rigid-support regime, we establish five gravitational results and delineate their exact relation to prior work: (i) symmetry-filtered spin-2 gravitational radiation from a $(6,4)$ Vernier cavity whose rigid C_6 and C_4 boundaries possess zero classical $l=2$ mass quadrupole ($\mathcal{I}^{\rm mass}_{ij}=2.72\times10^{-16}$), yielding dimensionless vacuum-sourced strain $\tilde{h}_+=0.1145$ ($0.3857\times$ relative to an equal-area periodic grating), alongside monopole ($l=0$) scalaron radiation ($\tilde{\varphi}_0=27.122$, Yukawa-suppressed by 9.80×10^{-4} above the Starobinsky mass threshold); (ii) an interior axial-force-free crossover at $\theta_0=25.3298^\circ$ ($|\tilde{F}_z|=0.0504$) where the coarse-grained curvature diagnostic remains finite ($\tilde{R}_{\rm eff}=-2317.8935$) while total apparatus mass remains strictly positive and asymptotic Tolman–Komar mass respects the Weak Equivalence Principle (WEP); (iii) a discrete-scale-invariant (DSI), log-periodic Newtonian potential and tidal curvature comb with complex scaling exponents $2+i(2\pi k/\ln b)$ ($\hat{\omega}_1=5.7096$, $\hat{b}=3.0055$); (iv) reversible rotary switching of local Null Energy Condition (NEC) and Strong Energy Condition (SEC) projections ($\widetilde{\rm NEC}_z\in[-579.895,6523.7421]$) and Raychaudhuri geodesic focusing bounded strictly within established sub-reflection worldline Quantum Energy Inequalities (QEIs, saturation ratio 0.1425); and (v) Cantor-stratified gravitomagnetic Lense–Thirring frame dragging ($\max|\tilde{\Omega}^{\rm vac}_{{\rm LT},z}|/\tilde{\Omega}=22931.2037$) sourced by broken vacuum boost invariance. Systematic prefractal generation ($N=1$–4), planar-limit (5.60×10^{-5}), and mesh-refinement tests confirm convergence of both the Casimir coefficient (0.0665) and its logarithmic scale derivative (0.072) within 0.0533 BEM discretization uncertainty.

I. INTRODUCTION AND LITERATURE REVIEW

The coupling between quantum vacuum fluctuations in quantum field theory (QFT) and the classical geometry of spacetime in general relativity (GR) lies at the heart of semiclassical gravity [1, 2],

$$G_{\mu\nu}+\Lambda g_{\mu\nu}\equiv R_{\mu\nu}-\frac{1}{2}g_{\mu\nu}R+\Lambda g_{\mu\nu}=\frac{8\pi G}{c^4}\langle\hat{T}_{\mu\nu}\rangle_{\rm ren},\quad(1)$$

where $\langle\hat{T}_{\mu\nu}\rangle_{\rm ren}$ denotes the renormalized expectation value of the quantum stress-energy tensor in a specified boundary and metric state, and we set the background cosmological constant $\Lambda=0$ in asymptotically flat Minkowski spacetime (noting, as emphasized by Cerdonio and Rovelli [3], that laboratory Casimir forces probe boundary-dependent vacuum energy differences rather than free-space Λ). Throughout this work we retain physical constants $(\hbar,c,G)$ explicitly in dimensional equations before mapping systematically to dimensionless Planck–Casimir variables, and adopt the metric signature $(-,+,+,+)$ [2, 4]. Since Casimir's discovery that conducting boundaries modify electromagnetic zero-point modes [5, 6], the Casimir apparatus has served as the canonical arena in which renormalized quantum

* Contact author: goren@gorengordon.com

stresses are finite, calculable from first principles, and non-classical. Yet most results on how Casimir vacua gravitate derive from maximally symmetric planar plates separated by a single length scale a, or treat the cavity as a passive test probe inside an externally prescribed curved metric.

A. Conformal Tracelessness, Trace Anomalies vs. Effective Traces, and Planar Boost Symmetry

In their foundational image-method calculation, Brown and Maclay [7] proved that for the source-free quantized electromagnetic field between ideal parallel conducting plates at $z = 0$ and $z = a$ (with interaction energy per unit area $E_C/A = -\pi^2\hbar c/(720a^3)$, equal to twice the single massless real scalar result $\epsilon_{\rm Cas}^{\rm scalar} = -\pi^2\hbar c/(1440a^4)$ due to two transverse photon polarizations [5, 6]), conformal invariance and in-plane (x, y) Poincaré symmetry uniquely fix the renormalized covariant and mixed stress-energy tensors in $(-,+,+,+)$ signature to

$$\begin{aligned}\langle\hat{T}_{\mu\nu}\rangle_{\rm flat} &= \frac{\pi^2\hbar c}{720a^4}\,{\rm diag}(-1,+1,+1,-3),\\ \langle\hat{T}^{\mu}{}_{\nu}\rangle_{\rm flat} &= {\rm diag}\big(-\rho_{\rm vac}c^2,\,P_\parallel,\,P_\parallel,\,P_\perp\big)\\ &= \frac{\pi^2\hbar c}{720a^4}\,{\rm diag}(+1,+1,+1,-3),\end{aligned}\tag{2}$$

where $T_{00} = +\rho_{\rm vac}c^2 = -\pi^2\hbar c/(720a^4) < 0$ is the negative vacuum energy density (so $T^0{}_0 = -\rho_{\rm vac}c^2 > 0$), $T_{xx} = T_{yy} = P_\parallel = -\rho_{\rm vac}c^2 > 0$ are the transverse tensions, and $T_{zz} = P_\perp = 3\rho_{\rm vac}c^2 < 0$ is the attractive normal pressure. Summing the mixed components confirms exact conformal tracelessness: $T^{\mu}{}_{\mu} = -\rho_{\rm vac}c^2 + 2P_\parallel + P_\perp = (1+1+1-3)\pi^2\hbar c/(720a^4) = 0$.

At the outset, it is essential to draw a sharp conceptual and gravitational distinction between three physically distinct trace concepts in quantum field theory and clarify how each couples to spacetime geometry:

1. *Local curved-spacetime conformal (Weyl) trace anomalies*: In curved backgrounds, ultraviolet regularization breaks classical Weyl invariance and generates a state-independent anomalous trace $\langle\hat{T}^{\mu}{}_{\mu}(x)\rangle_{\rm anom}$ proportional to geometric curvature invariants [1, 8]. As shown by Setare and Rezaeian [9] for Dirichlet plates in two-dimensional curved backgrounds, the local two-dimensional trace anomaly ($\propto R$) links boundary and background vacuum stresses directly; in $(3+1)$ dimensions, however, the local Weyl anomaly is quadratic in Riemann/Weyl curvature [8] and vanishes identically ($\mathcal{O}(\epsilon_P^4)$) in flat Minkowski interior vacuum $\mathbb{R}^3 \setminus \partial\Omega$.

2. *Non-conformal local operator traces*: For scalar fields with non-conformal curvature coupling $\xi \neq 1/6$ [10–12], bulk improvement terms $\propto (1/6 - \xi)\Box\langle\hat{\phi}^2(\mathbf{x})\rangle_{\rm ren}$ produce a genuine pointwise interior trace $\eta^{\mu\nu}\langle\hat{T}_{\mu\nu}(\mathbf{x})\rangle_{\rm ren} \neq 0$ even in flat space whenever $\langle\hat{\phi}^2(\mathbf{x})\rangle_{\rm ren}$ is inhomogeneous, sourcing a true pointwise interior Ricci scalar $R(\mathbf{x}) = -(8\pi G/c^4)\langle\hat{T}^{\mu}{}_{\mu}(\mathbf{x})\rangle_{\rm ren}$.

3. *Coarse-grained macroscopic effective trace and legitimacy as a gravitational source*: Recently, Gordon [13] proposed a phenomenological *unified effective trace ansatz* for self-similar Casimir hierarchies, arguing that when a cavity possesses an internal geometric length cutoff ℓ_* such that the dimensionless Casimir coefficient $\mathcal{C}(X)$ ($X \equiv \ln(d/\ell_*)$) runs log-periodically with plate separation d, the cavity-averaged macroscopic stress tensor acquires a non-vanishing *integrated effective trace* $T^{\mu}{}_{\mu,\rm eff} = -(\hbar c/d^4)\mathcal{C}'(X)$. Crucially, however, simply contracting Eq. (1) to $R = -(8\pi G/c^4)T^{\mu}{}_{\mu}$ and substituting a cavity-integrated ansatz $T^{\mu}{}_{\mu,\rm eff}$ is *not* equivalent to solving the local Einstein equations: first, contracting $G_{\mu\nu}$ discards the traceless Ricci and Weyl tensors that govern Newtonian potentials, tidal shears, gravitomagnetic frame dragging, and spin-2 waves; second, for conformally invariant fields ($\xi = 1/6$ and ideal electromagnetism), the pointwise local trace vanishes identically at every interior vacuum point $\mathbf{x} \in \mathbb{R}^3 \setminus \partial\Omega$ ($\eta^{\mu\nu}\langle\hat{T}_{\mu\nu}(\mathbf{x})\rangle_{\rm EM} = 0$, so $R_{\rm local}(\mathbf{x}) = 0$ in open interior vacuum), meaning that $T^{\mu}{}_{\mu,\rm eff}$ resides physically in distributional boundary/wall interaction stresses on $\partial\Omega$ and defines a *volume-averaged (coarse-grained) scalar curvature diagnostic* $R_{\rm eff} \equiv -(8\pi G/c^4)T^{\mu}{}_{\mu,\rm eff}$ over $V_{\rm cav}$—or a long-wavelength monopole ($l = 0$) source for the massive scalaron field φ in quadratic $f(R)$ gravity ($\lambda_\varphi \gg L$) when acoustic support relaxation lags behind external modulation. Accordingly, following the full linearized field-equation spirit of Buoninfante *et al.* [14] (who solved linearized quadratic-gravity field equations for parallel Casimir plates), we reconstruct the full local tensor $\langle\hat{T}_{\mu\nu}(\mathbf{x})\rangle_{\rm ren}$ throughout 3D space and solve the complete linearized tensor Einstein, Starobinsky $f(R)$, and gravitoelectromagnetic PDEs rather than relying on the scalar trace contraction alone.

Within linearized semiclassical gravity around flat spacetime, Eq. (2) imposes three structural degeneracies on canonical planar Casimir gravitation:

1. *Vanishing Ricci scalar and scalaron source*: Contracting Eq. (1) yields $R = -(8\pi G/c^4)\langle\hat{T}^{\mu}{}_{\mu}\rangle_{\rm ren} = 0$ across the planar gap, and constant $\mathcal{C} = -\pi^2/720$ gives $T^{\mu}{}_{\mu,\rm eff} = 0$. In quadratic $f(R) = R + \alpha R^2$ or scalar-tensor gravity [15–19], the extra scalar degree of freedom (scalaron) couples to the stress-energy trace and therefore decouples from planar conformal vacuum fluctuations.

2. *Algebraic locking of force, energy, and energy-condition violation*: Because $P_\perp = 3\rho_{\rm vac}c^2 < 0$, the planar force cannot vanish at finite separation without the vacuum energy also vanishing. Along the plate normal $k^\mu = (1,0,0,1)$, the Null Energy Condition (NEC) projection $T_{\mu\nu}k^\mu k^\nu = \rho_{\rm vac}c^2 + P_\perp = 4\rho_{\rm vac}c^2 < 0$ and Strong Energy Condition (SEC) combination $(T_{\mu\nu} - \frac{1}{2}g_{\mu\nu}T)u^\mu u^\nu = \frac{1}{2}(\rho_{\rm vac}c^2 + \sum_i P_i) = \rho_{\rm vac}c^2 < 0$ are unconditionally negative, forcing purely defocusing Raychaudhuri geodesic perturbations [20, 21].

3. *In-plane Lorentz boost invariance and rotational silence*: Parallel to the plates, the vacuum enthalpy density vanishes identically, $\rho_{\rm vac}c^2 + P_\parallel = 0$. Under rigid rotation of axisymmetric parallel disks relative to asymptotic inertial observers, the planar Casimir state acquires zero first-order laboratory momentum density $T^{0\phi}_{\rm vac} \propto v_\phi(\rho_{\rm vac}c^2 + P_\parallel) = 0$ (leaving only higher-order radiative vacuum friction $\propto \Omega^3$ on polarizable media [22, 23]), and therefore sources zero stationary gravitomagnetic Lense–Thirring field [24–27].

B. Prior Literature on Gravitational Casimir Physics, Dynamic Backgrounds, and Energy Inequalities

Extensive prior research has examined Casimir cavities placed inside externally prescribed gravitational fields. Early Fermi- and Rindler-coordinate analyses [28–30] showed that in a weak uniform gravitational field $Y \equiv g/c^2$ ($g_{00} \simeq -(1+2Yz)$), proper-time redshifting modifies the local scalar energy density across separation a as $\epsilon_{\rm Cas}(z) = -[\pi^2\hbar c/(1440a^4)][1+2Y(z-a/2)]$ (shifting by factor $1+2Ya$ across elevation a while leaving the symmetric spatial average unchanged to first order [28, 31]), and revealed apparent orientation-dependent forces when integrating $\langle\hat{T}_{\mu\nu}\rangle_{\rm ren}$ without mechanical support stresses. Fulling, Milton, Parashar, Romeo, Shajesh, and Wagner [32, 33] resolved this via variational principles, proving that covariant conservation $\nabla_\mu(T^{\mu\nu}_{\rm vac} + T^{\mu\nu}_{\rm con}) = 0$ causes spatial pressures to cancel in global mass integrals by the relativistic virial (Laue) theorem, yielding passive and active gravitational mass shift E_C/c^2 in compliance with the WEP. Crucially, Bekenstein [34] (via the subdominant trace energy condition) and Cerdonio and Rovelli [3, 35] proved that realistic plates and spring/mechanical supports always carry positive rest mass $M_{\rm plate} \gg |E_C|/c^2$, so total system mass $M_{\rm tot} = M_{\rm plate} + E_C/c^2 > 0$ is strictly positive and negative Casimir binding energy can never produce overall negative mass or gravitational levitation. Complementing the *Archimedes* vacuum-weighing experiment [36, 37], Sorge [38] applied the quasi-local Tolman and Komar mass formalisms to prove independently that a static Casimir cavity experiences a consistent WEP-compliant vacuum buoyancy shift E_C/c^2, while Sorge and Wilson [39, 40] demonstrated that a cavity in free fall toward a Schwarzschild black hole acquires extra energy from dynamical Casimir particle creation alongside tidal corrections.

More recently, time-dependent, rotating, and curved backgrounds have been shown to alter intracavity quantum states significantly. In flat Minkowski spacetime with rotating boundaries, Chernodub [41], Schaden [42], and Flachi and Edmonds [43] demonstrated that noninertial corotating frames modify quantum vacuum eigenmodes and Casimir free energies when tangential speeds approach relativistic scales or interact with background fields, though at zero temperature inside the speed-of-light cylinder ($\Omega L/c \ll 1$) no perpetual energy transfer occurs between zero-point fluctuations and rigid rotation [42] and intrinsic centrifugal/Coriolis shifts enter at $\mathcal{O}(\Omega^2L^2/c^2)$. In curved backgrounds, Sorge [44] demonstrated a *gravitational memory of the Casimir effect*, wherein a transient spacetime pulse leaves a permanent shift in vacuum energy; de Oliveira, Moreira, and Céleri [45] derived particle creation and parametric-amplification resonances in the dynamical Casimir effect (DCE) driven by passing gravitational waves; Sorge [46] and Zhang [47] proved that stationary Kerr-spacetime metric frame dragging (g_{0i}) renormalizes Casimir energy; and Buoninfante *et al.* [14], Pourhassan [19], and Garattini and Tzikas [48] analyzed Casimir cavities and wormholes in quadratic and $f(R) = R + \alpha R^2$ gravity. Crucially, Refs. [14, 44–47] address how external metrics or modified propagators perturb a passive cavity, whereas Refs. [19, 48] study exotic wormhole throats. Conversely, any negative Casimir energy interpretation is constrained by QEIs: Ford and Roman [49, 50] and Fewster and collaborators [51–53] proved that while a static Casimir vacuum violates unrestricted long-time averaged energy conditions, the difference-renormalized energy relative to empty Minkowski vacuum obeys smeared worldline bounds $\int_{-\infty}^{\infty}\langle\hat{T}_{00}(t)\rangle|f(t)|^2{\rm d}t \geq -C/\tau^4$ on sub-reflection causal diamonds not intersecting boundaries, preventing macroscopic causal pathologies while requiring positive boundary stress concentrations [11, 21, 54].

C. Fractal Casimir Precedents, Lapidus Boundary Spectral Theory, and Delineation of Novel Contributions

In flat Minkowski spacetime without gravity, it is crucial to distinguish *intrinsic fractal manifolds* or extra fractal compactified dimensions—where wave operators act on spaces of non-integer spectral or Hausdorff dimension [55–57]—from *Euclidean domains in* $\mathbb{R}^3$ *bounded by prefractal manifolds* $\partial\Omega$, governed mathematically by Lapidus's fractal-drum spectral theorem [58], in which bulk fields obey standard $(3+1)$-dimensional relativistic wave equations while self-similar boundaries imprint anomalous Weyl boundary terms and DSI log-periodic

spectral oscillations [59] analogous to Choptuik's critical gravitational collapse [60]. For self-similar plate hierarchies, Shajesh *et al.* [61, 62] solved explicit scalar recursion models for Cantor plate stacks and Sierpinski triangles, finding recursive sign changes and well-behaved convergence plateaus as prefractal order $N \to \infty$, while Gordon [13] proposed a phenomenological one-dimensional unified effective trace formula combining fractal thermodynamics with a postulated running Casimir coefficient. Independently, whereas mirror-symmetric bodies in vacuum always attract [63], Levin *et al.* [64] proved via exact BEM scattering [65–67] that an elongated conductor aligned with a metallic slot aperture in vacuum ($\varepsilon = 1$) experiences repulsive saddle-point axial Casimir force ($F_z > 0$).

To avoid any ambiguity between established results and new physics, we explicitly benchmark each of the five phenomena studied in our zero-temperature *Canonical Prefractal Sector Cavity* (CPSC) against prior literature (summarized also in Table I):

1. **Active Vernier Spin-2 Radiation and Positive-Energy $f(R)$ Scalaron Waves (Sec. IV — Uncharted Active Radiation Regime)**: Whereas de Oliveira *et al.* [45] and Sorge [44] study passive cavities driven by incoming GWs, we prove from cyclic scattering group theory (Appendix B) and Thorne's symmetric trace-free (STF) multipole expansion [68] that steady adiabatic rotation ($\Omega \ll \pi c/d_0$, zero photon creation) of a rigid C_6 rotor over a C_4 prefractal stator eliminates all classical rigid-body $l = 2$ mass-quadrupole emission while generating a rotating $m = 6 - 4 = 2$ vacuum energy wave that actively radiates spin-2 GWs at 6Ω, alongside ghost-free, positive-energy $l = 0$ Starobinsky scalaron waves ($\alpha > 0$) [18].

2. **Interior Force-Free Curvature Crossover Reconciled with Positive Total Mass and WEP (Sec. V)**: Consistent with Bekenstein [34], Cerdonio and Rovelli [3], Fulling *et al.* [32], and Sorge [38], our CPSC has strictly positive total apparatus mass ($M_{\rm plate} + E_C/c^2 > 0$, no anti-gravity levitation) and obeys the WEP asymptotically ($\Delta M_{\rm Komar}(\infty) = E_C(\theta)/c^2$). What is new is an *interior quasi-local decoupling* at the Levin-type repulsive-to-attractive crossover angle θ_0 [64]: vertical wall force vanishes ($F_z(\theta_0) = 0$) while logarithmic scale locking ($\partial_{\ln|z|}\mathcal{C} = 3\mathcal{C}$) preserves a finite interior coarse-grained curvature diagnostic $R_{\rm eff}(\theta_0) \neq 0$ and unscreened interior Tolman pressure integral inside $\tilde{r}_{\rm enc} < \tilde{r}_{\rm sup}$.

3. **3D Newtonian Metric and Quadratically Amplified Tidal Riemann Comb (Sec. VI)**: Whereas recursive Casimir scaling in flat space or extra fractal dimensions was studied in Refs. [13, 57, 61, 62], solving the 3D linearized Einstein Poisson equation shows how homothetic prefractal staging imprints complex scaling dimensions $2 + i(2\pi k/\ln b)$ onto the near-zone Newtonian potential h_{00} and amplifies log-periodic harmonics by $(2+i\omega_k)(3+i\omega_k)$ in the axial tidal Riemann tensor R_{0z0z}, with reversible rotary quenching.

4. **Reversible Rotary NEC/SEC Geodesic Switching Within Established Ford–Roman/Fewster QEIs (Sec. VII)**: We claim no new fractal QEI bound or violation of QFT energy inequalities; rather, working strictly within established Ford–Roman [49, 50] and Fewster–Pfenning [52, 53] difference QEIs on sub-reflection causal diamonds, we demonstrate reversible mechanical switching of the same axial slot channels between Raychaudhuri defocusing ($\widetilde{\rm NEC}_z < 0$) and converging microlensing ($\widetilde{\rm NEC}_z > 0, \widetilde{\rm SEC} > 0$).

5. **Active Vacuum Lense–Thirring Vorticity vs. Passive Kerr Frame Dragging (Sec. VIII)**: Whereas Sorge [46] and Zhang [47] analyzed passive Casimir cavities inside external Kerr metrics, we show within Costa and Natário's gauge-invariant GEM hierarchy [26, 27] that azimuthal slot walls break vacuum boost invariance ($\rho_{\rm vac}c^2 + P_\phi \neq 0$), sourcing active laboratory-frame momentum $T^{0\phi}_{\rm vac}$ and Cantor-stratified Lense–Thirring precession $\mathbf{\Omega}^{\rm vac}_{\rm LT}$.

The remainder of this article is organized as follows. Section II defines Planck–Casimir units, the CPSC geometry, the exact 1D/flat reduction of the 3D macroscopic effective trace, and covariant conservation. Section III details our boundary-element, four-pass voxel correlator, and spectral PDE solvers, together with prefractal ($N = 1$–4) and mesh-refinement convergence tests (Sec. III D). Sections IV–VIII present derivations and numerical benchmarks for the five effects. Section IX examines covariant consistency and physical scale estimates, and Sec. X summarizes our conclusions and caveats.

II. CANONICAL PREFRACTAL SECTOR CAVITY AND COVARIANT EFFECTIVE TRACE FORMALISM

A. Planck–Casimir Dimensionless Units and Dimensional Analysis

Rather than setting $\hbar = c = G = 1$ implicitly, we retain International System (SI) physical dimensions explicitly and verify dimensional homogeneity before mapping to dimensionless variables. Let L denote the outer cavity radius ($[L] = \mathrm{m}$). Zero-temperature Casimir energy densities carry dimensions $[\langle\hat{T}_{\mu\nu}\rangle_{\rm ren}] = [\hbar c/L^4] = \mathrm{J\,m^{-3}}$, while Einstein's gravitational coupling carries $[8\pi G/c^4] = \mathrm{m\,J^{-1}}$; their product has dimensions $[(8\pi G/c^4)\langle\hat{T}_{\mu\nu}\rangle_{\rm ren}] = \mathrm{m^{-2}} = [G_{\mu\nu}] = [R]$ and defines the

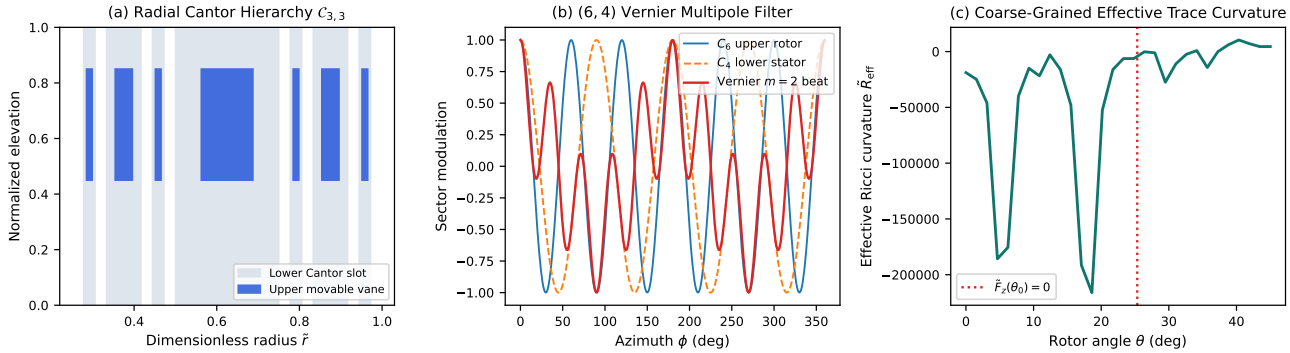


FIG. 1. Zero-temperature geometry and macroscopic effective-trace mechanism of the Canonical Prefractal Sector Cavity (CPSC) in dimensionless coordinates $\tilde{x}^\mu = x^\mu/L$ ($\epsilon_P = \ell_P/L$). (a) Analytic cross-sectional profile of the recursive middle-third ($b = 3, N = 3$) radial Cantor slot hierarchy $\mathcal{C}_{3,3}$ spanning $\tilde{r} \in [0.25, 1.00]$, showing lower stator aperture slots (gray) and centered upper rotor vanes (blue, fill ratio $\chi = 0.50$). (b) Analytic azimuthal multipole indicator modulation of the $(M_{\rm up}, M_{\rm low}) = (6, 4)$ Vernier constellation, illustrating how the product of C_6 rotor (solid blue) and C_4 stator (dashed orange) profiles synthesizes a pure $m = 2$ beat wave (red). (c) First-principles boundary-element numerical sweep of the dimensionless coarse-grained Ricci curvature $\tilde{R}_{\rm eff}(\theta) = -8\pi\tilde{T}^{\mu}{}_{\mu,\rm eff}(\theta)$ for ideal PEC electromagnetism across rotor angle $\theta \in [0°, 45°]$ in Constellation A $(4, 4)$, highlighting the axial-force-free crossover $\tilde{F}_z(\theta_0) = 0$ (vertical dotted line).

squared Planck length $\ell_P^2 = \hbar G/c^3 \approx 2.612 \times 10^{-70}$ m^2 and dimensionless Planck–cavity ratio $\epsilon_P \equiv \ell_P/L$. Similarly, in Starobinsky gravity $f(R) = R + \alpha R^2$, dimensional consistency of $f(R)$ requires $[\alpha] = [R^{-1}] = \text{m}^2$ ($\alpha > 0$), so the scalaron inverse Compton wavelength $m_\varphi = 1/\sqrt{6\alpha}$ carries [m^{-1}] (corresponding to rest mass $M_\varphi = \hbar/(c\sqrt{6\alpha})$) and $\tilde{m}_\varphi \equiv m_\varphi L = L/\sqrt{6\alpha}$ is dimensionless. To prevent floating-point underflow while preserving exact physical scaling, we define dimensionless tilde quantities:

$$\tilde{x}^\mu = \frac{x^\mu}{L}, \qquad \tilde{\kappa} = \kappa L, \qquad \tilde{E}_C = \frac{L}{\hbar c}E_C, \qquad \tilde{F}_z = \frac{L^2}{\hbar c}F_z,$$
$$\tilde{T}_{\mu\nu} = \frac{L^4}{\hbar c}\langle\hat{T}_{\mu\nu}\rangle_{\rm ren}, \qquad \tilde{R}_{\mu\nu} = \frac{L^4}{\ell_P^2}R_{\mu\nu}, \qquad \tilde{\bar{h}}_{\mu\nu} = \frac{L^2}{\ell_P^2}\bar{h}_{\mu\nu}, \tag{3}$$

where $\bar{h}_{\mu\nu} \equiv h_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}\eta^{\alpha\beta}h_{\alpha\beta}$ is the trace-reversed metric perturbation in Lorenz gauge $\partial^\mu\bar{h}_{\mu\nu} = 0$. In these units, the linearized semiclassical Einstein equation takes the parameter-free form

$$\tilde{\Box}\tilde{\bar{h}}_{\mu\nu}(\tilde{x}) = \left(-\frac{\partial^2}{\partial\tilde{x}_0^2} + \tilde{\nabla}^2\right)\tilde{\bar{h}}_{\mu\nu}(\tilde{x}) = -16\pi\,\tilde{T}^{\rm tot}_{\mu\nu}(\tilde{x}), \tag{4}$$

with local dimensionless Ricci scalar $\tilde{R}(\tilde{x}) = -8\pi\tilde{T}^{\mu}{}_{\mu,\rm tot}(\tilde{x})$. Physical SI values at any reference length scale L follow via $h_{\mu\nu} = \epsilon_P^2\tilde{h}_{\mu\nu}$ and $R = (\epsilon_P^2/L^2)\tilde{R}$.

B. Geometry of the Canonical Prefractal Sector Cavity

We consider two disjoint, rigid, ideally lossless boundary assemblies in $(3+1)$-dimensional Minkowski vacuum at $T = 0$ K spanning the cylindrical annulus $r \in [r_{\rm in}, r_{\rm out}] = [0.25L, 1.00L]$ ($\tilde{r} \in [0.25, 1.00]$, radial span $\Delta\tilde{r} = 0.75$; Fig. 1):

1. *Recursive radial Cantor hierarchy $\mathcal{C}_{N,b}$ and cutoff length ℓ_**: Given prefractal generation order $N \in \{0, 1, 2, 3, 4\}$ and scaling factor $b > 2$ (retention ratio $\eta = 1/b$ and central removal fraction $\gamma = 1 - 2/b$, yielding the middle-third Cantor set at $b = 3$), generation $m \in \{1, \ldots, N\}$ defines 2^{m-1} concentric aperture slots ($N_{\rm tracks} = 2^N - 1$ total slots for $N \geq 1$) of dimensionless radial width

$$\tilde{W}_m = \Delta\tilde{r}\left(1 - \frac{2}{b}\right)b^{-(m-1)}, \qquad m = 1, \ldots, N, \tag{5}$$

terminating at the physical short-distance prefractal cutoff length $\ell_* \equiv \Delta r\, b^{-N}$ (dimensionless $\tilde{\ell}_* = \ell_*/L = \Delta\tilde{r}\, b^{-N}$). Centered inside each slot $\tilde{r} \in [\tilde{r}_{c,k} - \tilde{W}_m/2, \tilde{r}_{c,k} + \tilde{W}_m/2]$ is an upper vane of width $\tilde{w}_m = \chi\tilde{W}_m$ ($\chi = 0.50$), yielding self-similar radial clearances $\tilde{g}_{r,m} = \frac{1}{2}(1-\chi)\tilde{W}_m = \tilde{g}_{r,1}b^{-(m-1)}$.

2. *Coplanar versus homothetic (r, z) vertical staging*: In the *coplanar* CPSC, lower aperture rims occupy $\tilde{z} \in [-\tilde{\tau}_0, 0]$ ($\tilde{\tau}_0 = 0.03$), upper vanes span $\tilde{z} \in [\tilde{z}_{\rm vane}, \tilde{z}_{\rm vane} + \tilde{h}_0]$ ($\tilde{h}_0 = 0.12$, baseline $\tilde{z}_{\rm vane} = -0.10$), and lower facing sectors sit at $\tilde{z}_{\rm top} = -0.12$ (vertical facing separation $\tilde{d}_0 = d_0/L = 0.02$). In the *homothetic (r, z)-staged* CPSC, cross-sectional dimensions and vertical aperture centers scale geometrically as

$$\tilde{z}_{{\rm stage},m} = -\tilde{Z}_1 b^{-(m-1)}, \qquad \tilde{h}_{{\rm rim},m} = \tilde{\tau}_1 b^{-(m-1)}, \tag{6}$$

with $\tilde{Z}_1 = 0.27$ and $\tilde{\tau}_1 = 0.054$, so rescaling $|\tilde{z}| \to |\tilde{z}|/b$ maps generation m onto generation $m+1$.

3. *Azimuthal multipole constellations* $(M_{\rm up}, M_{\rm low})$ *and controls*: Let $M_{\rm up}$ and $M_{\rm low}$ denote the discrete cyclic symmetry orders of the upper rotor and lower stator. Rotation by θ transitions vanes between open-aperture alignment ($\theta_{\rm open} = 0$, saddle-point fringing [64]) and facing-sector alignment ($\theta_{\rm face} = \pi/M_{\rm low}$, parallel-plate confinement [5, 7]). We analyze three constellations: (A) $(M_{\rm up}, M_{\rm low}) = (4, 4)$ breathing symmetry; (B) $(M_{\rm up}, M_{\rm low}) = (6, 4)$ Vernier quadrupole symmetry; and (C) chiral skew-sector $(4, 4)$ symmetry with azimuthal bevel $\delta\phi_m$. Each prefractal configuration is benchmarked against two iso-measure controls on $\tilde{r} \in [0.25, 1.00]$: a flat parallel annulus (`Ctrl_FlatAnnulus`) and an equal-slot-area single-scale periodic grating (`Ctrl_PeriodicGrating`) with $N_{\rm tracks}$ uniform slots of width $\tilde{W}_{\rm unif} = N_{\rm tracks}^{-1} \sum_k \tilde{W}_k$.

C. Quantum Field Models, Exact Reduction of the Macroscopic Effective Trace, and Covariant Conservation

To separate spin and local conformal properties, we evaluate three massless $T = 0$ K quantum field models on the CPSC boundaries $\partial\Omega = \partial\Omega_{\rm up} \cup \partial\Omega_{\rm low}$:

1. *Conformal massless real scalar* ($\xi = 1/6$): Governed by the Callan–Coleman–Jackiw improved stress-energy tensor [1, 10],

$$\hat{T}^{(\xi)}_{\mu\nu} = \partial_\mu\hat{\phi}\,\partial_\nu\hat{\phi} - \frac{1}{2}\eta_{\mu\nu}(\partial\hat{\phi})^2 + \xi\big(\eta_{\mu\nu}\Box - \partial_\mu\partial_\nu\big)\hat{\phi}^2, \quad (7)$$

satisfying $\eta^{\mu\nu}\langle\hat{T}^{(1/6)}_{\mu\nu}(\mathbf{x})\rangle_{\rm ren} = 0$ identically on shell ($\Box\hat{\phi} = 0$) everywhere in interior vacuum $\mathbb{R}^3 \setminus \partial\Omega$.

2. *Minimally coupled massless real scalar* ($\xi = 0$): Carries an intrinsic local bulk trace $\eta^{\mu\nu}\langle\hat{T}^{(0)}_{\mu\nu}(\mathbf{x})\rangle_{\rm ren} = -\frac{1}{2}\Box\langle\hat{\phi}^2(\mathbf{x})\rangle_{\rm ren} \neq 0$ whenever $\langle\hat{\phi}^2(\mathbf{x})\rangle_{\rm ren}$ varies spatially [11, 12] (attaining normalized interior magnitude 3.8948 in our CPSC gap).

3. *Ideal* 0 *K PEC vector electromagnetism*: Governed by the renormalized Maxwell stress-energy tensor

$$\langle\hat{T}_{00}\rangle_{\rm EM} = \frac{1}{2}\left\langle\varepsilon_0\mathbf{E}^2 + \mu_0\mathbf{H}^2\right\rangle_{\rm ren},$$
$$\langle\hat{T}_{ij}\rangle_{\rm EM} = \frac{1}{2}\delta_{ij}\left\langle\varepsilon_0\mathbf{E}^2 + \mu_0\mathbf{H}^2\right\rangle_{\rm ren} - \left\langle\varepsilon_0 E_i E_j + \mu_0 H_i H_j\right\rangle_{\rm ren}, \quad (8)$$

which is pointwise traceless ($\eta^{\mu\nu}\langle\hat{T}_{\mu\nu}(\mathbf{x})\rangle_{\rm EM} = 0$) at every interior point $\mathbf{x} \in \mathbb{R}^3 \setminus \partial\Omega$ (verified numerically in our CPSC solver to relative residual 1.25×10^{-16}).

To formulate the three-dimensional macroscopic effective trace unambiguously and connect phenomenological scaling [13] to first-principles prefractal scattering, define the dimensionless logarithmic separation and radial variables

$$X \equiv \ln\frac{d}{\ell_*}, \qquad Y \equiv \ln\frac{R}{\ell_*}, \quad (9)$$

at fixed physical prefractal cutoff length $\ell_* = \Delta r\, b^{-N}$ ($d = |z|$, annular area $A_{\rm ann} = \pi(r_{\rm out}^2 - r_{\rm in}^2) \propto R^2$). By dimensional analysis in $\hbar, c$, the exact zero-temperature Casimir interaction energy of a generation-N prefractal cavity writes without loss of generality as

$$\frac{E_{C,N}(\theta, d, R; \ell_*)}{A_{\rm ann}} = \frac{\hbar c}{d^3}\,\mathcal{C}_N(\theta, X, Y), \quad (10)$$

where $\mathcal{C}_{N,X} \equiv \partial\mathcal{C}_N/\partial X \equiv \mathrm{d}\mathcal{C}_N/\mathrm{d}(\ln d)|_{\theta,Y}$ and $\mathcal{C}_{N,Y} \equiv \partial\mathcal{C}_N/\partial Y \equiv \mathrm{d}\mathcal{C}_N/\mathrm{d}(\ln R)|_{\theta,X}$ denote explicit partial derivatives with respect to X and Y (written $\mathcal{C}'_N(X) \equiv \mathrm{d}\mathcal{C}_N/\mathrm{d}X$ when holding R fixed). Defining the coarse-grained vacuum energy density $\rho_{\rm vac}c^2 \equiv E_{C,N}/V_{\rm cav}$ and virtual-work macroscopic cavity pressures over $V_{\rm cav} = A_{\rm ann}d$ at fixed ℓ_* gives

$$\rho_{\rm vac}c^2 = \frac{\hbar c}{d^4}\mathcal{C}_N,$$
$$P_z = -\frac{1}{A_{\rm ann}}\frac{\partial E_{C,N}}{\partial d}\bigg|_{R,\ell_*} = \frac{\hbar c}{d^4}\big(3\mathcal{C}_N - \mathcal{C}_{N,X}\big),$$
$$P_r + P_\phi = -\frac{1}{d}\frac{R}{A_{\rm ann}}\frac{\partial E_{C,N}}{\partial R}\bigg|_{d,\ell_*} = -\frac{\hbar c}{d^4}\big(2\mathcal{C}_N + \mathcal{C}_{N,Y}\big). \quad (11)$$

Summing the mixed coarse-grained components $T^{\mu}{}_{\mu,\rm eff} \equiv -\rho_{\rm vac}c^2 + P_r + P_\phi + P_z$ cancels $3\mathcal{C}_N - \mathcal{C}_N - 2\mathcal{C}_N = 0$ identically, leaving in dimensional and dimensionless forms ($\tilde{d} = |\tilde{z}|$):

$$T^{\mu}{}_{\mu,\rm eff} = -\frac{\hbar c}{d^4}\big(\mathcal{C}_{N,X} + \mathcal{C}_{N,Y}\big)\Big|_{\ell_*},$$
$$\tilde{T}^{\mu}{}_{\mu,\rm eff}(\theta, \tilde{z}) = -\frac{1}{\tilde{V}_{\rm cav}}\mathcal{D}_{\rm cav}\tilde{E}_{C,N} = -\frac{1}{|\tilde{z}|^4}\big(\mathcal{C}_{N,X} + \mathcal{C}_{N,Y}\big)\Big|_{\tilde{\ell}_*}, \quad (12)$$

where $\mathcal{D}_{\rm cav} \equiv 1 + |\tilde{z}|\partial_{|\tilde{z}|}|_{\tilde{\ell}_*} + \tilde{R}\partial_{\tilde{R}}|_{\tilde{\ell}_*}$. In the 1D plate-stack limit ($\mathcal{C}_{N,Y} = 0$), Eq. (12) reduces identically to $T^{\mu}{}_{\mu,\rm eff} = -(\hbar c/d^4)\mathcal{C}'_N(X)$ [13]; for featureless plates ($\mathcal{C} = -\pi^2/720$), $T^{\mu}{}_{\mu,\rm eff} = 0$; and by Euler homogeneity, $\mathcal{D}_{\rm cav}\tilde{E}_{C,N} = -\tilde{\ell}_*\partial_{\tilde{\ell}_*}\tilde{E}_{C,N}$ reflects boundary work against the prefractal cutoff $\tilde{\ell}_*$. Moreover, whereas finite-temperature radiation on an intrinsic fractal of spectral dimension d_s obeys Akkermans, Dunne, and Teplyaev's fractal Stefan–Boltzmann equation of state $p_{\rm th}V_s = U_{\rm th}/d_s$ [55]—yielding coarse-grained thermal trace $T^{\mu}{}_{\mu,\rm th} = -\rho_{\rm th} + 3p_{\rm th} = \rho_{\rm th}(3/d_s - 1)$ per spectral volume V_s [13]—our CPSC operates strictly at $T = 0$ K

($\rho_{\rm th}=0$) in Euclidean $\mathbb{R}^3$ bulk [58], so $T^{\mu}{}_{\mu,\rm th}=0$ identically.

Crucially, in Ref. [13], postulating a log-periodic $\mathcal{C}(X)$ constituted a phenomenological ansatz rather than a microscopic derivation. Here we both derive $\mathcal{C}_N(X)$ analytically for a self-similar prefractal model and compute $\mathcal{C}_N$ from first principles via Eq. (16). Specifically, in a homothetic generation-N Cantor hierarchy with characteristic stage depths $Z_m = Z_1 b^{-(m-1)}$ ($m=1,\dots,N$, cutoff $\ell_* = Z_1 b^{-N}$, $X_0 \equiv \ln(d/Z_1) = X - N\ln b$), each self-similar stage contributes through a localized dimensionless Euclidean Yukawa scattering window $\Phi(\xi)$ ($\Phi(\xi)\to 0$ for $|\xi|\gg 1$), obeying the exact finite-N recursion

$$\mathcal{C}_N(X_0) = \mathcal{C}_{\rm bg} + \sum_{m=1}^{N} q^{m-1}\,\Phi\big(X_0 + (m-1)\ln b\big), \quad (13)$$

where q is the inter-generation coupling weight. At fixed physical gap $d>0$ in a coplanar Cantor cavity, sub-wavelength generations ($W_m \ll d$) are Yukawa-screened with $q<1$ ($q_{\rm area}=2/b=2/3$ geometrically and $q_{\rm evan}=2/b^2=2/9$ for evanescent dipole scattering), guaranteeing absolute convergence of both $\lim_{N\to\infty}\mathcal{C}_N(d)$ and $\lim_{N\to\infty}\mathcal{C}'_N(X)$. Conversely, probing homothetically across the scaling window $\ell_* \ll d \ll Z_1$ ($q=1$ along the self-similar ray) and applying Poisson summation as N grows yields the discrete-scale-invariant Fourier–Mellin expansion

$$\mathcal{C}_N(\theta,X) \simeq \sum_{k=-\infty}^{\infty} \mathcal{C}_k(\theta) e^{i\omega_k X}, \qquad \omega_k = \frac{2\pi k}{\ln b}, \quad \mathcal{C}_k = \frac{\hat{\Phi}(\omega_k)}{\ln b}, \quad (14)$$

establishing how discrete finite-N modes transition smoothly into log-periodic running $\mathcal{C}'_N(X) \simeq \sum_{k\neq 0} i\omega_k \mathcal{C}_k e^{i\omega_k X}$ inside $\ell_* \ll d \ll Z_1$ while vanishing outside the prefractal window.

Finally, Lorenz-gauge Einstein consistency ($\partial^\mu \tilde{\Box}\tilde{\bar{h}}_{\mu\nu}=0$) requires exact conservation of the *total* stress-energy tensor:

$$\partial_\mu \tilde{T}^{\mu\nu}_{\rm tot} = \partial_\mu\Big(\tilde{T}^{\mu\nu}_{\rm vac} + \tilde{T}^{\mu\nu}_{\rm con}\Big) = 0, \quad (15)$$

where $\tilde{T}^{\mu\nu}_{\rm con}$ represents mechanical support stresses balancing Casimir forces $\tilde{F}_z(\theta)$ and torques $\tilde{\tau}_\theta(\theta) = -\partial_\theta \tilde{E}_C(\theta)$ [32, 33, 38], plus shaft power $\partial_{\tilde{t}}\tilde{E}_C = -\tilde{\Omega}\tilde{\tau}_\theta$. Throughout, we state three explicit physical idealizations: (i) boundaries and supports are modeled in the rigid, lossless limit ($Y_{\rm rigid}\to\infty$, elastic energy $\sim P^2/Y_{\rm rigid}\to 0$); (ii) rotation is strictly sub-relativistic inside the light cylinder ($\tilde{\Omega}=\Omega L/c \ll 1$), so non-inertial Unruh/corotating vacuum shifts [41–43] are suppressed to $\mathcal{O}(\tilde{\Omega}^2)$; and (iii) while classical plate rest mass $M_{\rm plate}$ and moment of inertia $I_{\rm plate}\Omega$ dominate static monopole backgrounds in hardware, C_6/C_4 symmetry cancels the classical $l=2$ mass quadrupole identically in Phenomenon I while angular/radial differential subtraction isolates the vacuum backreaction in Phenomena II–V (Appendix A).

III. FIRST-PRINCIPLES SCATTERING, VOXEL CORRELATOR, AND CONVERGENCE VERIFICATION

To evaluate $\tilde{E}_C$, $\tilde{T}_{\mu\nu}(\tilde{\mathbf{x}})$, $\tilde{T}^{\mu}{}_{\mu,\rm eff}$, and all induced gravitational perturbations without proximity-force approximations, we implement three coupled solvers verified by systematic convergence tests.

A. 3D Cyclic-Bloch Boundary-Element Scattering, Prefractal Regularization, and Scale-Derivative Engine

Along the full imaginary frequency axis $\xi \in (-\infty,+\infty)$, the zero-temperature Casimir interaction energy reads $E_C = \frac{\hbar}{2}\int_{-\infty}^{\infty}\frac{d\xi}{2\pi}\ln\det[\mathbf{I}-\mathbf{M}(|\xi|)]$; folding onto the positive Wick wavenumber axis $\tilde{\kappa}=\xi L/c\in(0,\infty)$ cancels the zero-point $\frac{1}{2}$ factor universally [65, 66]:

$$\tilde{E}_C(\theta,\tilde{z}) = \frac{1}{2\pi}\int_0^\infty d\tilde{\kappa}\,\ln\det\big[\mathbf{I}-\mathbf{M}(\tilde{\kappa};\theta,\tilde{z})\big], \quad (16)$$

where polarization multiplicity is encoded directly in the dimension of $\mathbf{M}(\tilde{\kappa}) = \mathbf{\Gamma}_{\rm UU}^{-1}\mathbf{\Gamma}_{\rm UL}\mathbf{\Gamma}_{\rm LL}^{-1}\mathbf{\Gamma}_{\rm LU}$ ($N_p\times N_p$ for one real scalar field, reproducing $\tilde{\epsilon}^{\rm scalar}_{\rm Cas} = -\pi^2/(1440\tilde{d}^4)$ in the planar limit; $2N_p\times 2N_p$ for two-polarization PEC electromagnetism, reproducing $\tilde{\epsilon}^{\rm EM}_{\rm Cas} = -\pi^2/(720\tilde{d}^4)$).

A key theoretical requirement in prefractal geometries is rigorous ultraviolet regularization and renormalization of edge and corner divergences. In single-domain zeta-function or Seeley–DeWitt heat-kernel expansions [6, 69], $\mathrm{Tr}(e^{-s\Delta})$ generates bulk volume (s^{-2}), Weyl surface area ($s^{-3/2}$), 90° dihedral slot-edge ($s^{-1}L_{\rm edge}\propto 2^N$), and corner ($s^{-1/2}$) divergences alongside finite prefractal crossover boundary counterterms [13, 58, 61]. Because our CPSC consists of two strictly disjoint rigid assemblies ($\partial\Omega_{\rm U}\cap\partial\Omega_{\rm L}=\emptyset$ with minimum clearance $\tilde{d}_{\rm min}>0$), every local bulk, surface, 2^N-edge, and corner divergence resides inside the isolated self-energies of $\partial\Omega_{\rm U}$ or $\partial\Omega_{\rm L}$ alone. Factoring the coupled scattering determinant as $\det\mathbf{\Gamma}_{\rm U+L}/(\det\mathbf{\Gamma}_{\rm UU}\det\mathbf{\Gamma}_{\rm LL}) = \det[\mathbf{I}-\mathbf{M}(\tilde{\kappa})]$ in Eq. (16) cancels all bulk, surface, prefractal edge, and corner counterterms identically prior to $\tilde{\kappa}$-integration, leaving $\mathbf{M}(\tilde{\kappa})$ trace-class and exponentially convergent ($\|\mathbf{M}(\tilde{\kappa})\|\sim e^{-2\tilde{\kappa}\tilde{d}_{\rm min}}$ as $\tilde{\kappa}\to\infty$) without ad hoc counterterms. Boundary surfaces are meshed into N_p quadrilateral/annular panels $\{\mathcal{P}_a\}_{a=1}^{N_p}$ with piecewise-constant basis functions; integrable panel self-interactions are evaluated analytically via equal-area disk regularization $\tilde{\sigma}_a = \sqrt{\tilde{A}_a/(16\pi)}$:

- *Scalar Dirichlet kernel*: Single-layer Euclidean Yukawa kernel $G_0(\tilde{\mathbf{r}},\tilde{\mathbf{r}}';\tilde{\kappa}) = \exp(-\tilde{\kappa}|\tilde{\mathbf{r}}-\tilde{\mathbf{r}}'|)/(4\pi|\tilde{\mathbf{r}}-\tilde{\mathbf{r}}'|)$ (N_p degrees of freedom).

- *PEC vector-dyadic kernel*: Electric-field integral equation (EFIE) projecting two tangential surface-current basis vectors $(\hat{\mathbf{t}}_{a,1}, \hat{\mathbf{t}}_{a,2})$ per panel onto the Euclidean dyadic Green's tensor $\mathbf{G}(\tilde{\mathbf{r}}, \tilde{\mathbf{r}}'; \tilde{\kappa}) = \big(\mathbf{I} - \tilde{\kappa}^{-2}\tilde{\nabla}\tilde{\nabla}\big)G_0(\tilde{\mathbf{r}}, \tilde{\mathbf{r}}'; \tilde{\kappa})$ ($2N_p$ degrees of freedom).

For constellations with cyclic greatest common divisor $M_g = \gcd(M_{\rm up}, M_{\rm low}) > 1$ ($M_g = 4$ for $(4,4)$ and $M_g = 2$ for $(6,4)$), discrete Fourier projection into M_g decoupled Bloch blocks $\mathbf{M}^{(\ell)}(\tilde{\kappa}) = \sum_{q=0}^{M_g-1} e^{2\pi i \ell q/M_g}\mathbf{M}_{0,q}(\tilde{\kappa})$ accelerates LU factorization by M_g^3 without approximation (verified against full-ring dense LU to $< 10^{-11}$ relative error). Integrals over $\tilde{\kappa}$ use log-mapped Gauss–Legendre quadrature on $\tilde{\kappa} \in [10^{-2}, 1.5\times 10^2]$, and parametric derivatives ($\tilde{F}_z$, $\tilde{\tau}_\theta$, $\mathcal{D}_{\rm cav}\tilde{E}_C$) use centered finite differences of logarithmic step $\Delta \ln d = 0.015$.

B. Four-Pass Inclusion–Exclusion 3D Euclidean Correlator Voxel Mapping

To reconstruct the spatially resolved renormalized stress tensor $\tilde{T}_{\mu\nu}(\tilde{\mathbf{x}};\theta)$ at interior vacuum points $\tilde{\mathbf{x}} \in \Omega_{\rm grid} \subset \mathbb{R}^3 \setminus \partial\Omega$ (masked outside solid conductor volumes by safety clearance $\tilde{\delta}_{\rm wall} = 0.008$, corresponding physically to finite plasma skin-depth regularization [11, 54]), we evaluate the scattered Euclidean Green's correlator $\Delta G(\tilde{\mathbf{x}}, \tilde{\mathbf{x}}'; \tilde{\kappa}) \equiv G^{\rm full}(\tilde{\mathbf{x}}, \tilde{\mathbf{x}}'; \tilde{\kappa}) - G_0(\tilde{\mathbf{x}}, \tilde{\mathbf{x}}'; \tilde{\kappa})$. Subtracting free-space coincidence divergences (G_0) directly and cancelling single-body edge/surface stresses via four-pass inclusion–exclusion yields

$$\Delta\tilde{T}^{\rm mut}_{\mu\nu}(\tilde{\mathbf{x}};\theta) = \tilde{T}^{\rm U+L}_{\mu\nu}(\tilde{\mathbf{x}};\theta) - \tilde{T}^{\rm U}_{\mu\nu}(\tilde{\mathbf{x}};\theta) - \tilde{T}^{\rm L}_{\mu\nu}(\tilde{\mathbf{x}}) + \tilde{T}^{\rm vac}_{\mu\nu}(\tilde{\mathbf{x}}), \tag{17}$$

alongside the boundary-renormalized total interior tensor $\tilde{T}^{\rm ren}_{\mu\nu}(\tilde{\mathbf{x}};\theta) = \tilde{T}^{\rm U+L}_{\mu\nu}(\tilde{\mathbf{x}};\theta) - \tilde{T}^{\rm vac}_{\mu\nu}(\tilde{\mathbf{x}})$, evaluated from the LU-factorized scattering inverse $\mathbf{\Gamma}^{-1}(\tilde{\kappa})$:

$$\Delta G_{\rm ren}(\tilde{\mathbf{x}}, \tilde{\mathbf{x}}'; \tilde{\kappa}) = -\sum_{a,b} g_0(\tilde{\mathbf{x}}, \mathcal{P}_a; \tilde{\kappa})\big[\mathbf{\Gamma}^{-1}(\tilde{\kappa})\big]_{ab} g_0(\mathcal{P}_b, \tilde{\mathbf{x}}'; \tilde{\kappa}). \tag{18}$$

Analytical spatial derivatives $\partial_i\partial'_j \Delta G_{\rm ren}|_{\tilde{\mathbf{x}}'=\tilde{\mathbf{x}}}$ and $\tilde{\kappa}^2 \Delta G_{\rm ren}(\tilde{\mathbf{x}}, \tilde{\mathbf{x}}; \tilde{\kappa})$ assemble the scalar (7) and dyadic electromagnetic (8) stress tensors at each voxel.

C. 3D Spectral Linearized Einstein, Scalaron, and Gravitomagnetic PDE Solvers

All linear-algebra, quadrature, Fast Fourier Transform (FFT), and root-finding routines are implemented in double-precision Python using NumPy [70] and SciPy [71]. From $\tilde{T}^{\rm tot}_{\mu\nu}(\tilde{\mathbf{x}}, \tilde{t})$, gravitational observables are computed across three sectors:

1. *Far-zone radiative multipole integration*: Whereas purely static prefractal configurations ($\partial_{\tilde{t}}\mathcal{C}_N = 0$) source only stationary near-zone potentials without gravitational radiation, time-dependent rotor modulation $\theta(\tilde{t}) = \tilde{\Omega}\tilde{t}$ couples to radiative STF mass and current multipoles [68]. In the slow-motion regime ($\tilde{\omega} = \omega L/c \ll 1$), transverse-traceless spin-2 waveforms along $\hat{\mathbf{n}} = (\sin\vartheta\cos\varphi, \sin\vartheta\sin\varphi, \cos\vartheta)$ follow Thorne's conserved $l = 2$ mass-quadrupole formula [2, 4, 68],

$$\tilde{h}^{\rm TT}_{ij}(\tilde{t}, \tilde{r}\hat{\mathbf{n}}) = \frac{2}{\tilde{r}}\Lambda_{ij,kl}(\hat{\mathbf{n}})\frac{\mathrm{d}^2\tilde{\mathcal{Q}}^{\rm tot}_{kl}}{\mathrm{d}\tilde{t}^2}(\tilde{t} - \tilde{r}),$$
$$\tilde{\mathcal{Q}}^{\rm tot}_{kl}(\tilde{t}) = \int \tilde{T}^{\rm tot}_{00}(\tilde{\mathbf{x}}, \tilde{t})\left(\tilde{x}_k\tilde{x}_l - \frac{1}{3}\tilde{\mathbf{x}}^2\delta_{kl}\right)\mathrm{d}^3\tilde{x}, \tag{19}$$

cross-checked against $2\int(\tilde{T}^{\rm tot}_{kl})^{\rm TF}\mathrm{d}^3\tilde{x}$ via Appendix A. In linearized $f(R) = R + \alpha R^2$ gravity ($\alpha > 0$, dimensionless scalaron mass $\tilde{m}_\varphi = L/\sqrt{6\alpha}$), Fourier modes $\tilde{\mathcal{T}}_{\rm eff}(\tilde{\omega})$ of the trace monopole evolve via the exact retarded/Yukawa Klein–Gordon Green's function [18]:

$$\tilde{\varphi}(\tilde{t}, \tilde{r}) = -\frac{2\alpha_\varphi}{\tilde{r}}\int_0^\infty \mathrm{d}\tilde{\omega}\,\mathrm{Re}\Big[\tilde{\mathcal{T}}_{\rm eff}(\tilde{\omega})\,\mathcal{G}_\varphi(\tilde{\omega}, \tilde{r})e^{-i\tilde{\omega}\tilde{t}}\Big],$$
$$\mathcal{G}_\varphi(\tilde{\omega}, \tilde{r}) = \begin{cases} e^{i\sqrt{\tilde{\omega}^2 - \tilde{m}_\varphi^2}\,\tilde{r}}, & \tilde{\omega} > \tilde{m}_\varphi, \\ e^{-\sqrt{\tilde{m}_\varphi^2 - \tilde{\omega}^2}\,\tilde{r}}, & \tilde{\omega} \le \tilde{m}_\varphi, \end{cases} \tag{20}$$

with canonical coupling $\alpha_\varphi = 1/\sqrt{6}$.

2. *Near-zone 3D isolated-boundary Poisson solver*: Elliptic equations $\tilde{\nabla}^2\tilde{\bar{h}}_{\mu\nu} = -16\pi\tilde{T}^{\rm tot}_{\mu\nu}$ and $\tilde{\nabla}^2\tilde{h}_{0i} = +16\pi\tilde{T}^{0i}_{\rm tot}$ with open falloff $\tilde{\bar{h}}_{\mu\nu} \to 0$ as $|\tilde{\mathbf{x}}| \to \infty$ are solved via $2\times$ zero-padded Hockney–Eastwood 3D FFT convolution with the spherical-equivalent regularized kernel $1/(4\pi|\tilde{\mathbf{x}} - \tilde{\mathbf{x}}'|)$ (reproducing analytical uniform-ball potentials within $< 3\%$), using fourth-order central stencils for $\tilde{R}_{0z0z} = -\frac{1}{2}\partial^2_{\tilde{z}}\tilde{h}_{00}$ and $\tilde{\mathbf{B}}^{\rm vac}_g = \tilde{\nabla}\times\tilde{\mathbf{h}}_0$.

3. *Raychaudhuri and QEI quadrature*: Geodesic expansion shifts and Lorentzian-weighted worldline QEI integrals are evaluated by adaptive quadrature [71] along axial rays and stationary azimuthal observer worldlines.

D. Regularization, Truncation Error Control, Planar Limit, and Prefractal Convergence ($N \to \infty$)

To ensure reproducibility and verify physical limits as well as $N \to \infty$ convergence [13, 61], Fig. 2 and our consistency suite establish four benchmarks:

1. *Convergence of $\mathcal{C}_N$, $\mathcal{C}'_N(X)$, and $\tilde{T}^{\mu}{}_{\mu,{\rm eff},N}$ ($N = 1,2,3,4$)*: At fixed gap $\tilde{d}_0 = 0.02$, once $\tilde{W}_N \lesssim \tilde{d}_0$, evanescent Yukawa modes ($\tilde{\kappa} \sim 1/\tilde{d}_0$) smooth over finer sub-slots so both the Casimir coefficient and its logarithmic derivative plateau geometrically (Fig. 2a,b): across $N = 3 \to 4$, $|\Delta\mathcal{C}_N|/|\mathcal{C}_N|$

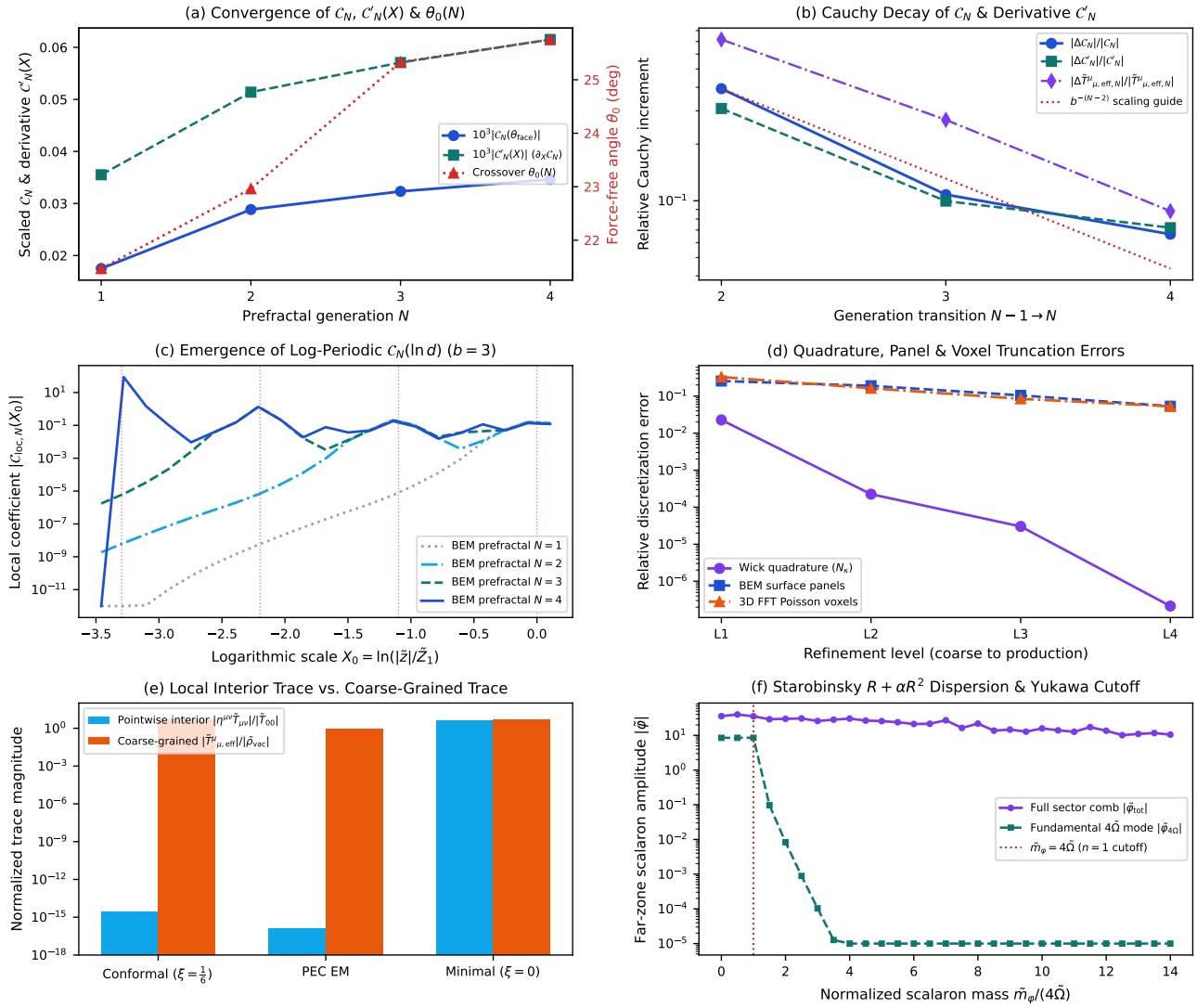


FIG. 2. Numerical verification of prefractal convergence ($N = 1$–4), log-periodic $\mathcal{C}_N(\ln d)$ emergence, local-vs.-macroscopic trace separation, discretization error bounds, and Starobinsky $f(R)$ dispersion in dimensionless Planck–Casimir units. (a) Fixed-gap ($\tilde{d}_0 = 0.02$) BEM convergence of the dimensionless Casimir coefficient $10^3|\mathcal{C}_N(\theta_{\rm face})|$ (solid circles), its logarithmic derivative $10^3|\mathcal{C}'_N(X)|$ ($X = \ln(\tilde{d}/\tilde{\ell}_*)$, dashed squares), and the force-free crossover angle $\theta_0(N)$ (dotted triangles, right axis) across $N = 1, 2, 3, 4$ ($b = 3$). (b) Semilogarithmic Cauchy decay of $|\Delta\mathcal{C}_N|/|\mathcal{C}_N|$, $|\Delta\mathcal{C}'_N|/|\mathcal{C}'_N|$, and $|\Delta\tilde{T}^{\mu}{}_{\mu,\mathrm{eff},N}|/|\tilde{T}^{\mu}{}_{\mu,\mathrm{eff},N}|$ versus transition $N-1 \to N$ against the analytical $b^{-(N-2)}$ scaling guide (dotted red). (c) First-principles BEM emergence of log-periodic oscillations in the local scale-compensated Casimir coefficient $|\mathcal{C}_{\mathrm{loc},N}(X_0)| = |\tilde{z}|^4|\Delta\tilde{T}^{\rm mut}_{00}|$ swept across $X_0 = \ln(|\tilde{z}|/\tilde{Z}_1)$ for $N = 1, 2, 3, 4$ ($b = 3$; vertical dotted lines mark stage centers $-(m-1)\ln 3$). (d) Monotonic decay of truncation errors across refinement levels L1–L4 for Gauss–Legendre Wick quadrature ($N_\kappa \in \{6, 10, 14, 20\}$ vs. 28), isotropic BEM surface panels, and 3D FFT Poisson voxel grids. (e) Direct numerical comparison of pointwise interior trace $|\eta^{\mu\nu}\tilde{T}_{\mu\nu}|/|\tilde{T}_{00}|$ (blue bars, vanishing to $\sim 10^{-16}$ for $\xi = 1/6$ and PEC EM, finite for $\xi = 0$) versus coarse-grained cavity trace $|\tilde{T}^{\mu}{}_{\mu,\mathrm{eff}}|/|\tilde{\rho}_{\mathrm{vac}}|$ (orange bars). (f) Far-zone Starobinsky $f(R) = R + \alpha R^2$ ($\alpha > 0$) scalaron amplitude $|\tilde{\varphi}|$ versus $\tilde{m}_\varphi/(4\tilde{\Omega})$ comparing the full sector-edge comb $|\tilde{\varphi}_{\mathrm{tot}}|$ against the fundamental $4\tilde{\Omega}$ mode $|\tilde{\varphi}_{4\Omega}|$.

drops to 0.0665, the logarithmic derivative increment $|\mathcal{C}'_4 - \mathcal{C}'_3|/|\mathcal{C}'_4|$ drops to 0.072, the 3D effective trace increment drops to 0.0881, and the force-free angle stabilizes at $\theta_0(N = 4) = 25.7528°$ (shift $0.423°$ from $\theta_0(3) = 25.3298°$).

2. *Emergence of log-periodic $\mathcal{C}_N(\ln d)$ and local-vs.-macroscopic trace verification*: Sweeping $X_0 = \ln(|\tilde{z}|/\tilde{Z}_1)$ across the homothetic CPSC for $N = 1, 2, 3, 4$ confirms Eq. (13) directly (Fig. 2c): each added generation N appends one new log-periodic lobe at $-(N-1)\ln 3$ while leaving coarser stages converged. Simultaneously, Fig. 2e confirms that

pointwise interior traces vanish to machine precision for $\xi = 1/6$ and PEC EM (1.25×10^{-16}) while remaining $\mathcal{O}(1)$ locally for $\xi = 0$ (3.8948) and coarse-grained over $V_{\rm cav}$ for all three fields.

3. *Non-fractal planar Casimir limit and flat-space stress-energy conservation*: In the featureless parallel-plate limit, our log-mapped Wick quadrature reproduces the exact single-scalar Casimir density $-\pi^2/(1440\tilde{d}^4)$ (and PEC EM density $-\pi^2/(720\tilde{d}^4)$) to relative error 5.60×10^{-5}, while fourth-order interior spatial stencils confirm flat-space vacuum conservation $|\partial_j \tilde{T}^{ji}_{\rm vac}|/\|\tilde{\nabla}\tilde{T}_{\rm vac}\| = 1.68 \times 10^{-10}$ across all three QFT models.

4. *Wick quadrature, BEM panel, and 3D voxel truncation uncertainties*: At $N_\kappa = 14$, the quadrature error against $N_\kappa = 28$ is 3.02×10^{-5}. Isotropic BEM panel refinement up to $(5, 5, 3)$ yields discretization uncertainty 0.0533, and 3D FFT GEM voxel refinement up to $48 \times 48 \times 32$ yields vorticity uncertainty 0.0529 (shown as $\pm\sigma_{\rm BEM}$ and $\pm\sigma_{\rm vox}$ bands in Figs. 4 and 7).

IV. PHENOMENON I: SYMMETRY-FILTERED SPIN-2 VERNIER GRAVITATIONAL RADIATION AND PERTURBATIVE $f(R)$ SCALARON WAVES

A. Elimination of Classical Rigid-Body Quadrupole Background and Vernier Selection Rule

Consider a rigid rotor of classical mass density $\rho_{\rm mass}(r, \phi, z)$ invariant under M-fold discrete rotations $\phi \to \phi + 2\pi/M$. Expanding $\rho_{\rm mass} = \sum_{n\in\mathbb{Z}} \rho_{nM}(r, z)e^{inM\phi}$, its symmetric trace-free (STF) $l = 2, m = \pm 2$ mass quadrupole moments [68] $I^{\rm mass}_{\pm 2} = \int \rho_{\rm mass}(r, \phi, z)r^2 e^{\pm 2i\phi} r\,{\rm d}r\,{\rm d}\phi\,{\rm d}z$ vanish identically whenever $M \geq 3$, because $nM = \pm 2$ admits no integer solution $n \in \mathbb{Z}$ for $M \geq 3$. Consequently, rotating a rigid C_6 vane assembly at constant angular velocity Ω above a stationary C_4 stator produces identically zero classical $l = 2$ mass-quadrupole gravitational radiation ($\dddot{I}^{{\rm mass},C_6}_{ij} = 0$, verified on our 3D mesh to residual 2.72×10^{-16}); in Thorne's STF multipole hierarchy [68], classical rigid-plate emission is suppressed to the $l = 6$ duodecapole by an additional relativistic factor $(\Omega L/c)^8$ (and in Constellation A $(4, 4)$, $m = 0, 4$ symmetry eliminates $l = 2$ spin-2 waves completely).

As proved directly from cyclic group representations of the surface-scattering operator $\mathbf{M}(\tilde{\kappa}; \theta)$ in Appendix B, the renormalized Casimir interaction energy density $\Delta\tilde{T}^{\rm vac}_{00}(\tilde{r}, \phi, \tilde{z}; \theta)$ couples the $M_{\rm up} = 6$ rotor harmonics and $M_{\rm low} = 4$ stator harmonics nonlinearly, generating azimuthal wave numbers $m = pM_{\rm up} - qM_{\rm low}$ $(p, q \in \mathbb{Z})$. At lowest non-trivial order $(p, q) = (\pm 1, \pm 1)$, this yields $m = \pm(6 - 4) = \pm 2$:

$$\Delta\tilde{T}^{\rm vac}_{00}(\tilde{r}, \phi, \tilde{z}; \tilde{t}) = \bar{\mathcal{E}}_0(\tilde{r}, \tilde{z}) + 2\,{\rm Re}\Big[\mathcal{E}_2(\tilde{r}, \tilde{z})\, e^{i(2\phi - 6\tilde{\Omega}\tilde{t})}\Big] + \ldots, \tag{21}$$

where $\tilde{\Omega} \equiv \Omega L/c \ll 1$. Physically, sectors at $\phi = 0, \pi$ occupy the attractive facing phase ($\Delta\tilde{T}^{\rm vac}_{00} < 0$) while orthogonal sectors at $\phi = \pm\pi/2$ occupy the open-aperture saddle-point phase, sweeping an $m = 2$ quantum-vacuum energy wave at pattern speed $\Omega_{\rm pat} = 3\Omega$ while the angle-integrated energy modulates only at ${\rm lcm}(6, 4)\Omega = 12\Omega$ (zero net shaft torque at 6Ω). Crucially, because $6\Omega \ll \pi c/d_0$, the cavity evolves adiabatically in its instantaneous ground state without exciting dynamical Casimir photons; conversely, as shown in Appendix B, placing the stationary $(6, 4)$ cavity in an incoming external GW recovers the parametric resonance structure of de Oliveira *et al.* [45] and GW work coupling of Sorge [44] when $\omega_{\rm GW} = 6\Omega$. Substituting Eq. (21) into Eq. (19) yields circularly polarized on-axis ($\vartheta = 0$) spin-2 waveforms at $\omega_{\rm GW} = 6\Omega$:

$$\begin{aligned}
\tilde{h}_+(\tilde{t}, \tilde{r}, \vartheta) &= -\frac{72\,\tilde{\Omega}^2 |\tilde{Q}^{(6\Omega)}_2|}{\tilde{r}} \frac{1 + \cos^2\vartheta}{2} \cos\big(6\tilde{\Omega}(\tilde{t} - \tilde{r}) - \delta_2\big), \\
\tilde{h}_\times(\tilde{t}, \tilde{r}, \vartheta) &= -\frac{72\,\tilde{\Omega}^2 |\tilde{Q}^{(6\Omega)}_2|}{\tilde{r}} \cos\vartheta \sin\big(6\tilde{\Omega}(\tilde{t} - \tilde{r}) - \delta_2\big),
\end{aligned} \tag{22}$$

with dimensionless Peters–Mathews power $\tilde{P}^{(2,2)}_{\rm GW} = \frac{46656}{5}\tilde{\Omega}^6 |\tilde{Q}^{(6\Omega)}_2|^2$ (physical luminosity $P_{\rm GW} \propto G\hbar^2\Omega^6 L^2/c^7$), attaining $\tilde{h}_+ = 0.1145$ for PEC EM and 0.1888 for $\xi = 1/6$ at $N = 3$, versus 0.2969 for the single-scale periodic control (ratio 0.3857× due to radial Cantor area partitioning; Fig. 3b).

B. Perturbative Field Equations and Yukawa Dispersion of the $f(R)$ Scalaron

To formulate the scalar breathing mode rigorously [15, 17–19], consider quadratic Starobinsky gravity $f(R) = R + \alpha R^2$ with $\alpha > 0$ (ensuring both ghost freedom $f'(0) = 1 > 0$ and Dolgov–Kawasaki tachyon stability $f''(0) = 2\alpha > 0$). Varying the metric action yields the exact modified field equations [17, 19]

$$f'(R)R_{\mu\nu} - \frac{1}{2}f(R)g_{\mu\nu} + \big(g_{\mu\nu}\Box - \nabla_\mu\nabla_\nu\big)f'(R) = \frac{8\pi G}{c^4}T^{\rm tot}_{\mu\nu}, \tag{23}$$

whose trace $3\Box f'(R) + f'(R)R - 2f(R) = (8\pi G/c^4)T^{\mu}{}_{\mu,{\rm tot}}$ linearizes around Minkowski spacetime ($R^{(0)} = 0$) in terms of the canonically normalized scalaron $\varphi \equiv \sqrt{3/2}\,\delta f'(R) = \sqrt{6}\,\alpha\,\delta R$ ($\alpha_\varphi = 1/\sqrt{6}$) with effective mass $m_\varphi = 1/\sqrt{6\alpha} > 0$ ($\tilde{m}_\varphi = L/\sqrt{6\alpha}$) to the Klein–Gordon wave equation

$$\big(\tilde{\Box} - \tilde{m}^2_\varphi\big)\,\tilde{\varphi}(\tilde{t}, \tilde{\mathbf{x}}) = 8\pi\alpha_\varphi\,\tilde{T}^{\mu}{}_{\mu,{\rm tot}}(\tilde{t}, \tilde{\mathbf{x}}). \tag{24}$$

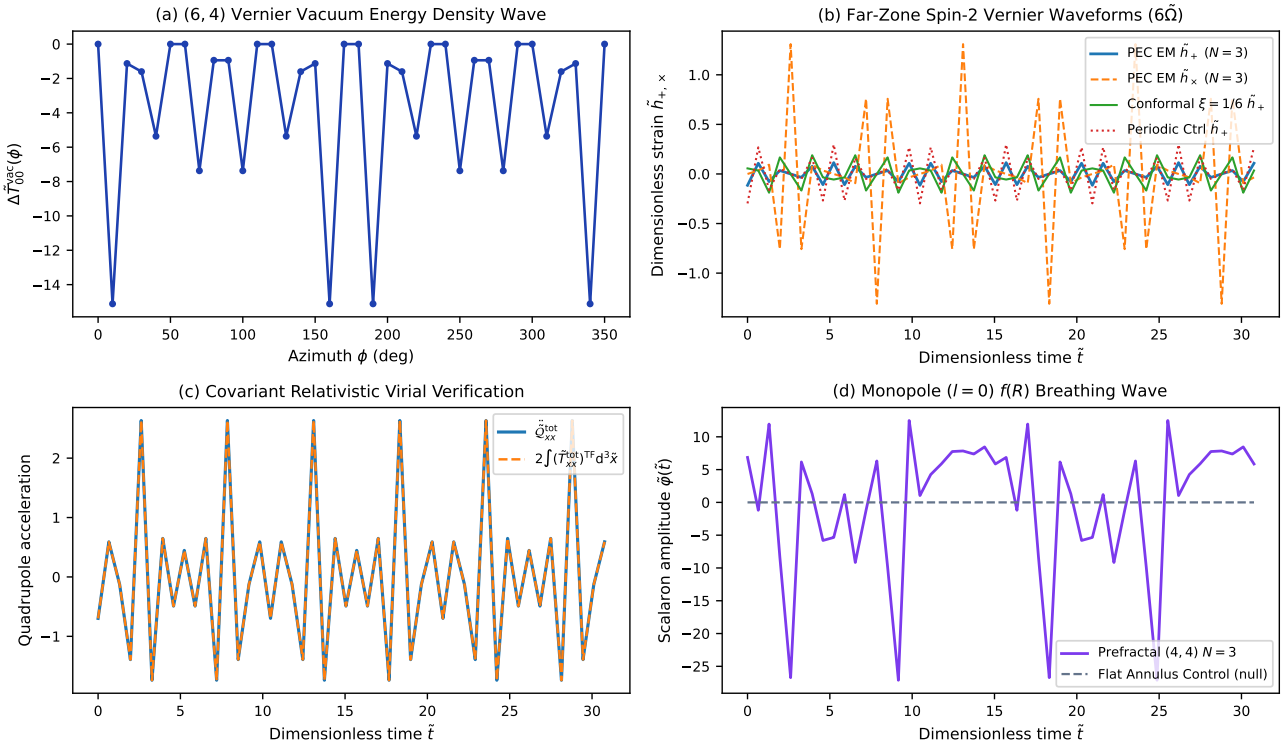


FIG. 3. Symmetry-filtered tensor and scalar-tensor gravitational radiation generated by steady adiabatic rotation ($\tilde{\Omega} = 0.10, \tilde{r}_{\rm obs} = 10$) of the CPSC in dimensionless units ($\epsilon_P = \ell_P/L$). (a) Numerical four-pass BEM reconstruction of the azimuthal Casimir vacuum energy wave $\Delta\tilde{T}^{\rm vac}_{00}(\phi)$ in the (6, 4) Vernier cavity ($N = 3$), exhibiting a pure $m = 2$ spatial quadrupole while classical rigid C_6 and C_4 mass quadrupoles vanish to machine precision (2.72×10^{-16}). (b) Numerical far-zone spin-2 waveforms $\tilde{h}_+(\tilde{t})$ and $\tilde{h}_\times(\tilde{t})$ oscillating at $\tilde{\omega}_{\rm GW} = 6\tilde{\Omega}$ for PEC EM (solid blue $\tilde{h}_+$, dashed orange $\tilde{h}_\times$), conformal scalar $\xi = 1/6$ (solid green), and equal-area periodic grating control (dotted red). (c) Numerical verification of the relativistic second-moment virial theorem (A2), comparing $\ddot{\tilde{Q}}^{\rm tot}_{xx}$ (solid) against $2\int(\tilde{T}^{\rm tot}_{xx})^{\rm TF}\mathrm{d}^3\tilde{x}$ (dashed; relative error 1.45×10^{-15}). (d) Monopole ($l = 0$) Starobinsky $f(R)$ breathing waveform $\tilde{\varphi}(\tilde{t})$ in the massless propagation limit for the (4, 4) prefractal cavity ($N = 3$, purple) versus the identically silent flat-annulus analytical/numerical control (dashed gray).

By the trace of the virial identity (A2), if an exterior mechanical support frame were infinitely rigid with instantaneous relativistic sound speed, static boundary reaction stresses would cancel $\int_{\rm cav}\tilde{P}^{\rm vac}_i\mathrm{d}^3\tilde{x}$; however, for any physical support ring whose elastic sound speed $c_s \ll c$ places its acoustic relaxation rate $\tau_{\rm el}^{-1} \sim c_s/L$ below the 4Ω breathing modulation of Constellation A (4, 4) (where external drive power $\dot{\tilde{E}}_C = -\tilde{\Omega}\tilde{\tau}_\theta(\tilde{t})$ pumps the cavity energy), the uncompensated oscillatory cavity trace $\tilde{\mathcal{T}}_{\rm eff}(\tilde{t}) = \int_{\rm cav}\tilde{T}^{\mu}{}_{\mu,\rm eff}\mathrm{d}^3\tilde{x} = \sum_{n=1}^{\infty}\tilde{\mathcal{T}}_{4n}\cos(4n\tilde{\Omega}\tilde{t})$ sources Eq. (24) via Eq. (20). Moreover, because $f'(0) = 1 > 0$ and $\alpha > 0$, Berry and Gair's Isaacson wave-zone expansion of Eq. (23) [18] guarantees strictly positive-definite radiated energy density and outward flux ($t^{\rm eff}_{0r} = \frac{c^4}{32\pi G}\langle(\partial_t\bar{h}^{\rm TT}_{ij})^2\rangle - c\langle\partial_t\varphi\,\partial_r\varphi\rangle \geq 0$), extracted physically from the external mechanical drive. In the flat-annulus control, $\tilde{T}^{\mu}{}_{\mu,\rm eff} = 0$ (numerical waveform amplitude 0). In the (4, 4) prefractal CPSC ($N = 3$), $\tilde{\varphi}$ attains amplitude 27.122 in the light-scalaron limit ($\tilde{m}_\varphi \ll 4\tilde{\Omega}$; Fig. 3d), whereas for a heavy Starobinsky scalaron ($\tilde{m}_\varphi = 2\times4\tilde{\Omega}$, small $\alpha > 0$) the fundamental $4\tilde{\Omega}$ mode becomes evanescent and decays at $\tilde{r}_{\rm obs} = 10$ to $9.80\times10^{-4}\times$ its massless amplitude, while sharp sector-edge harmonics $4n\tilde{\Omega}$ peel off sequentially across the angular-transit comb before total Yukawa screening sets in (Fig. 2f).

V. PHENOMENON II: FORCE-FREE SPACETIME CURVATURE CROSSOVER AND TOLMAN–KOMAR ACTIVE MASS

In a stationary asymptotically flat spacetime possessing timelike Killing vector $\xi^\mu = (1, \mathbf{0})$ and unit hypersurface normal n^μ, the general-relativistic Komar active gravitational mass enclosed within radius $r_{\rm enc}$ reduces at linear order ($\sqrt{-g} \simeq 1$, $g_{00} \simeq -1$) in $(-,+,+,+)$ signa-

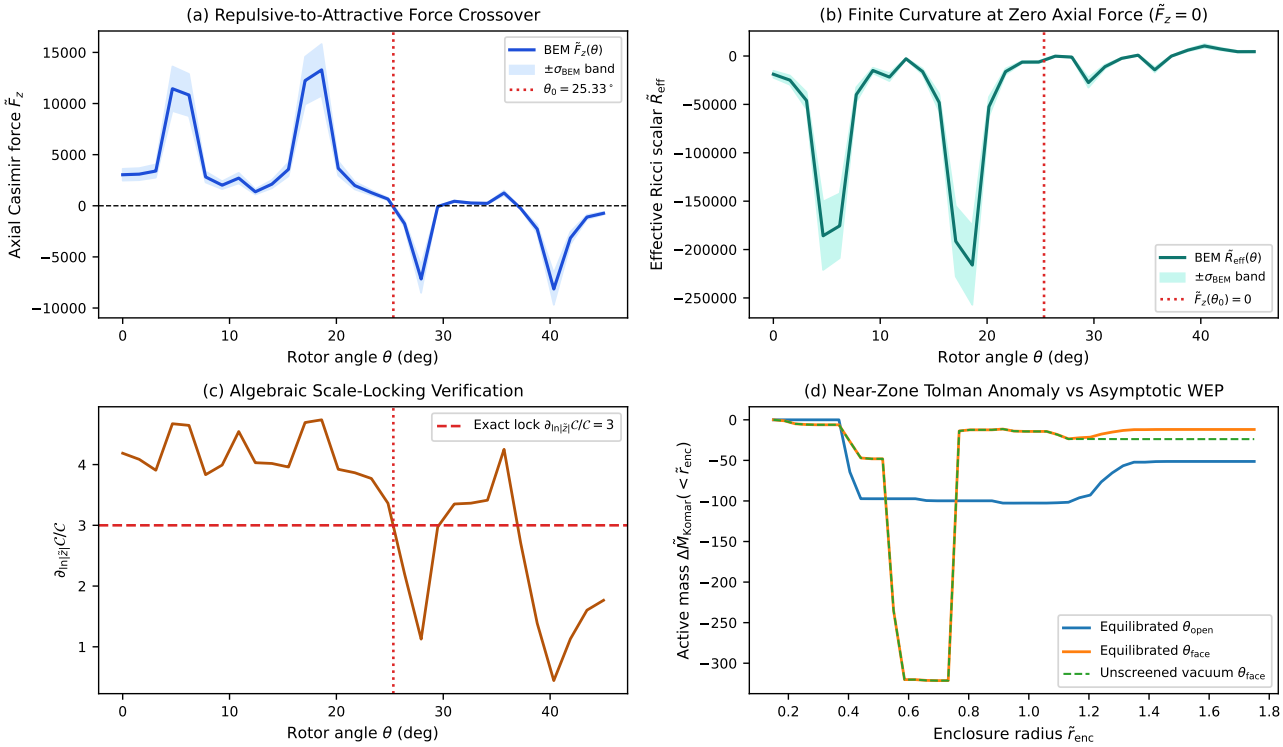


FIG. 4. Decoupling of axial mechanical boundary force from coarse-grained spacetime curvature and radial resolution of the Tolman–Komar active gravitational mass in Constellation A $(4,4)$ ($N = 3$, dimensionless units $\epsilon_P = \ell_P/L$). (a) Numerical BEM sweep of dimensionless axial Casimir force $\tilde{F}_z(\theta)$ with $\pm\sigma_{\rm BEM}$ discretization uncertainty band (shaded blue), crossing continuously from open-aperture repulsion ($\tilde{F}_z > 0$) to facing-sector attraction ($\tilde{F}_z < 0$) at $\theta_0 = 25.3298^\circ$ (vertical dotted line). (b) Numerical coarse-grained Ricci curvature $\tilde{R}_{\rm eff}(\theta) = -8\pi\tilde{T}^{\mu}{}_{\mu,\rm eff}(\theta)$ with $\pm\sigma_{\rm BEM}$ error band (shaded teal), remaining non-zero (-2317.8935) at $\tilde{F}_z(\theta_0) = 0$. (c) Numerical verification of the analytical scale-locking condition $\partial_{\ln|\tilde{z}|}\mathcal{C}/\mathcal{C} = 3$ (dashed red horizontal line) at $\theta = \theta_0$. (d) Cumulative radial integral of the active gravitational mass $\Delta\tilde{M}_{\rm Komar}(<\tilde{r}_{\rm enc})$ from Eq. (25), contrasting unscreened interior vacuum stresses (dashed green) against self-equilibrated cavities supported at $\tilde{r}_{\rm sup} = 1.25$ in open (blue) and facing (orange) states, recovering $\Delta\tilde{M}_{\rm Komar}(\infty) = \tilde{E}_C(\theta)$ asymptotically.

ture to Tolman's active mass formula [2, 4, 38, 72, 73]:

$$\begin{aligned}\Delta\tilde{M}_{\rm Komar}(\tilde{r}_{\rm enc};\theta) &= 2\int_{|\tilde{\mathbf{x}}|<\tilde{r}_{\rm enc}}\left(\tilde{T}^{\rm tot}_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}\tilde{T}^{\alpha}{}_{\alpha,\rm tot}\right)\\ &\quad\times n^\mu\xi^\nu \mathrm{d}^3\tilde{x}\\ &= \int_{|\tilde{\mathbf{x}}|<\tilde{r}_{\rm enc}}\left(2\tilde{T}^{\rm tot}_{00} + \tilde{T}^{\mu}{}_{\mu,\rm tot}\right)\mathrm{d}^3\tilde{x}\\ &= \int_{|\tilde{\mathbf{x}}|<\tilde{r}_{\rm enc}}\left(\tilde{\rho}_{\rm tot} + \sum_{i=x,y,z}\tilde{P}_{i,\rm tot}\right)\mathrm{d}^3\tilde{x}.\end{aligned} \tag{25}$$

For a complete, static, self-equilibrated cavity-plus-support system ($\partial_j\tilde{T}^{ij}_{\rm tot} = 0$ with support ring inside $\tilde{r}_{\rm sup}$), total apparatus rest mass $M_{\rm tot} = M_{\rm plate} + \tilde{E}_C(\theta)/c^2 > 0$ is strictly positive [3, 34, 35], and the identity $\tilde{T}^{ij}_{\rm tot} = \partial_k(\tilde{x}^i\tilde{T}^{kj}_{\rm tot})$ guarantees $\int_{\mathbb{R}^3}\tilde{P}_{i,\rm tot}\,\mathrm{d}^3\tilde{x} = 0$, recovering the differential binding-mass shift $\Delta\tilde{M}_{\rm Komar}(\infty;\theta) = \tilde{E}_C(\theta)$ in exact agreement with Fulling *et al.* [32] and Sorge [38]. Within the pre-fractal cavity, however, two distinct interior gravitational features emerge:

1. *Force-free curvature crossover ($\theta = \theta_0$)*: Between open-aperture repulsion $\tilde{F}_z(0,\tilde{z}) > 0$ [64] and facing-sector attraction $\tilde{F}_z(\theta_{\rm face},\tilde{z}) < 0$, continuity guarantees an angle $\theta_0(\tilde{z})$ where net axial force vanishes:

$$\tilde{F}_z(\theta_0,\tilde{z}) = -\left.\frac{\partial\tilde{E}_C}{\partial|\tilde{z}|}\right|_{\theta_0} = \frac{\tilde{A}_{\rm ann}}{|\tilde{z}|^4}\left[3\mathcal{C}(\theta_0) - \partial_{\ln|\tilde{z}|}\mathcal{C}(\theta_0)\right] = 0. \tag{26}$$

At $\theta_0 = 25.3298^\circ$ (force residual $|\tilde{F}_z| = 0.0504$), the axial support constraint vanishes completely while $\tilde{E}_C(\theta_0) < 0$ remains finite, locking $\partial_{\ln|\tilde{z}|}\mathcal{C}|_{\theta_0} = 3\mathcal{C}(\theta_0)$ (numerical ratio 3.0018). Substituting into Eq. (12) yields a finite *force-free coarse-grained Ricci curvature*:

$$\tilde{R}_{\rm eff}(\theta_0,\tilde{z}) = -8\pi\tilde{T}^{\mu}{}_{\mu,\rm eff}(\theta_0,\tilde{z}) = \frac{8\pi}{|\tilde{z}|^4}\left[3\mathcal{C}(\theta_0) + \partial_{\ln\tilde{R}}\mathcal{C}(\theta_0)\right], \tag{27}$$

attaining $\tilde{R}_{\rm eff}(\theta_0) = -2317.8935$ at $N = 3$ (Fig. 4b,c).

2. *Quasi-local interior Tolman pressure integral versus asymptotic WEP mass*: When the mechanical support ring sits at exterior radius $\tilde{r}_{\rm sup} = 1.25 > \tilde{r}_{\rm out}$, a quasi-local integration surface $\tilde{r}_{\rm enc} \in [\tilde{r}_{\rm in}, \tilde{r}_{\rm out}]$ encloses the uncompensated interior vacuum pressure $\int_{\rm cav} \sum_i \tilde{P}_{i,\rm vac} {\rm d}^3\tilde{x} \neq 0$, whereas integrating past $\tilde{r}_{\rm sup}$ cancels spatial stresses and yields a pure WEP binding-mass shift $\Delta\tilde{M}_{\rm Komar}(\infty) = \tilde{E}_C(\theta_{\rm face}) - \tilde{E}_C(0) = 39.4077$ matching the near-zone equilibrated swing 39.4065 to 3×10^{-5} at fixed baryon number (Fig. 4d).

VI. PHENOMENON III: DISCRETE-SCALE-INVARIANT LOG-PERIODIC SPACETIME METRIC COMB

In critical gravitational collapse, Choptuik [60] showed that discrete self-similarity produces log-periodic echoing in spacetime curvature corresponding to complex scaling exponents [56, 59]. In our homothetically staged CPSC (6), static Lorenz-gauge Einstein theory gives $\tilde{\nabla}^2\tilde{h}_{00}(\tilde{\mathbf{x}}) = -8\pi\big(\tilde{\rho}_{\rm tot} + \sum_i \tilde{P}_{i,\rm tot}\big) = -8\pi\big(2\tilde{\rho}_{\rm tot} + \tilde{T}^{\mu}{}_{\mu,\rm tot}\big)$. Along the collision-free homothetic open-aperture slot-center ray $\tilde{\mathbf{x}}(\tilde{z}) = \big(\tilde{r}_{\rm in} + \frac{1}{2}\Delta\tilde{r}(|\tilde{z}|/\tilde{Z}_1), 0, -1.044|\tilde{z}|\big)$ with upper vane elevation $\tilde{z}_{\rm vane} = -0.956|\tilde{z}|$, rescaling $|\tilde{z}| \to |\tilde{z}|/b$ maps the scattered Green's function of generation m onto generation $m+1$ across $\tilde{\ell}_* \ll |\tilde{z}| \ll \tilde{Z}_1$. Mellin transformation of the Newtonian Green's integral $\tilde{h}_{00}(\tilde{\mathbf{x}};\tilde{z}) = 2\int|\tilde{\mathbf{x}}-\tilde{\mathbf{x}}'|^{-1}(2\tilde{\rho}_{\rm tot} + \tilde{T}^{\mu}{}_{\mu,\rm tot}){\rm d}^3\tilde{x}'$ yields complex scaling exponents $\Delta_k = 2 + i\omega_k$ ($\omega_k = 2\pi k/\ln b$):

$$\tilde{h}_{00}(\tilde{\mathbf{x}};\theta_{\rm open},\tilde{z}) = \frac{1}{|\tilde{z}|^2}\sum_{k=-\infty}^{\infty}\mathcal{H}_k^{(00)}\left(\frac{\tilde{\mathbf{x}}}{|\tilde{z}|}\right)\exp\left(i\frac{2\pi k}{\ln b}\ln\frac{|\tilde{z}|}{\tilde{\ell}_*}\right), \tag{28}$$

while the axial tidal Riemann component $\tilde{R}_{0z0z} = -\frac{1}{2}\partial_{\tilde{z}}^2\tilde{h}_{00}$ scales as $|\tilde{z}|^{-4}$ with quadratically amplified harmonic weights $(2+i\omega_k)(3+i\omega_k)\mathcal{H}_k^{(00)}$. Rotating to $\theta_{\rm face}$ engages facing-sector confinement and quenches the sidewall comb. Across $N = 4$ homothetic Cantor stages ($b = 3$), harmonic regression on $\delta\ln(|\tilde{z}|^2|\tilde{h}_{00}|)$ recovers $\hat{\omega}_1 = 5.7096$ and $\hat{b} = 3.0055$ (0.17% and 0.18% from exact $2\pi/\ln 3 = 5.7192$ and $b = 3$, respectively; $R^2 = 0.6301$ vs. 0.2411 in the periodic control, peak open tidal curvature 1585.3059, and rotary harmonic suppression 3.1177×; Fig. 5).

VII. PHENOMENON IV: ROTARY ENERGY-CONDITION SWITCHING, RAYCHAUDHURI GEODESIC LENSING, AND WORLDLINE QEIS

At linear order in ϵ_P^2, test null congruences ($k^\mu = (1,\hat{\mathbf{k}})$) and slow timelike congruences ($u^\mu = (1,\mathbf{0})$) with zero initial twist and shear obey the Raychaudhuri equations [2, 20, 21]:

$$\frac{{\rm d}\tilde{\Theta}_{\rm null}}{{\rm d}\tilde{\lambda}} = -\tilde{R}_{\mu\nu}k^\mu k^\nu = -8\pi\underbrace{\big(\tilde{T}_{00} + \tilde{T}_{ij}\hat{k}^i\hat{k}^j\big)}_{\widetilde{\rm NEC}_{\hat{\mathbf{k}}}}, \tag{29}$$

$$\frac{{\rm d}\tilde{\Theta}_{\rm time}}{{\rm d}\tilde{\tau}} = -\tilde{R}_{\mu\nu}u^\mu u^\nu = -4\pi\underbrace{\big(2\tilde{T}_{00} + \tilde{T}^{\mu}{}_{\mu}\big)}_{\widetilde{\rm SEC}}. \tag{30}$$

Whereas null focusing is blind to $\tilde{T}^{\mu}{}_{\mu}$ ($g_{\mu\nu}k^\mu k^\nu = 0$), timelike focusing couples directly to $\tilde{T}^{\mu}{}_{\mu}$, distinguishing $\xi = 1/6$ from $\xi = 0$ via $-\frac{1}{2}\tilde{\Box}\langle\hat{\phi}^2\rangle_{\rm ren}$:

1. *Rotary switching between defocusing and focusing*: At $\theta_{\rm face}$, mode exclusion gives $\tilde{T}_{00} < 0$ and $\tilde{T}_{zz} \approx 3\tilde{T}_{00} < 0$, so $\widetilde{\rm NEC}_z$ reaches -579.895 and axial rays defocus ($\mathrm{d}\tilde{\Theta}_{\rm null}/\mathrm{d}\tilde{\lambda} > 0$). At $\theta_{\rm open}$, transverse slot fringing ($\langle E_r^2\rangle > \langle E_z^2\rangle$) renders $\tilde{T}_{zz} = \tilde{P}_z > |\tilde{\rho}_{\rm vac}| > 0$, flipping $\widetilde{\rm NEC}_z$ up to 6523.7421 and $\widetilde{\rm SEC} \geq 0$ across volume fraction 0.9959, yielding net null expansion swing 100.8036 and trace-to-net timelike focusing ratio 40.2844 at $\xi = 0$ (Fig. 6a–c).

2. *Sub-reflection worldline QEIs in bounded domains with ideal conductors*: Reconciling negative Casimir densities with classic QEI theorems requires care [49, 52, 53]: as established by Ford and Roman [49], static Casimir vacua violate unrestricted long-time ($\tau \to \infty$) averaged energy conditions because $-C/\tau^4 \to 0$, whereas difference-renormalized expectation values relative to empty vacuum satisfy short-time QEIs; moreover, idealized Dirichlet/PEC boundaries induce surface-divergent densities $\propto d_{\rm wall}^{-4}$ as $d_{\rm wall} = {\rm dist}(\mathbf{x},\partial\Omega) \to 0$ [11] (regularized in real metals at the plasma skin depth λ_p [54]). By Fewster and Pfenning's local-covariance theorem for globally hyperbolic subregions [52, 53], difference QEIs relative to free Minkowski vacuum hold without boundary-reflection terms—and with the standard universal constant $C_{\rm FR}$ unmodified by boundary fractality—for sampling functions localized inside causal diamonds that do not intersect $\partial\Omega$ ($\tau < d_{\rm wall}/c$, enforcing $|\rho_{\rm vac}| \lesssim C_{\rm FR}\hbar c/d_{\rm wall}^4$). A stationary gap observer at $(\tilde{r}_0,\phi_0,\tilde{z}_0)$ (wall clearance $\tilde{d}_{\rm wall} = \tilde{d}_0/2 > \tilde{\delta}_{\rm wall}$) sampling $\tilde{T}_{00}(\tilde{t}) = \tilde{T}_{00}(\tilde{r}_0,\phi_0-\tilde{\Omega}\tilde{t},\tilde{z}_0)$ over sub-reflection Lorentzian width $\tilde{\tau} = (\tilde{\tau}_\Omega^{-2} + \tilde{d}_{\rm wall}^{-2})^{-1/2} < \tilde{d}_{\rm wall}$ ($\tilde{\tau}_\Omega = \Delta\phi_{\rm vane}/(M_{\rm up}\tilde{\Omega})$) therefore

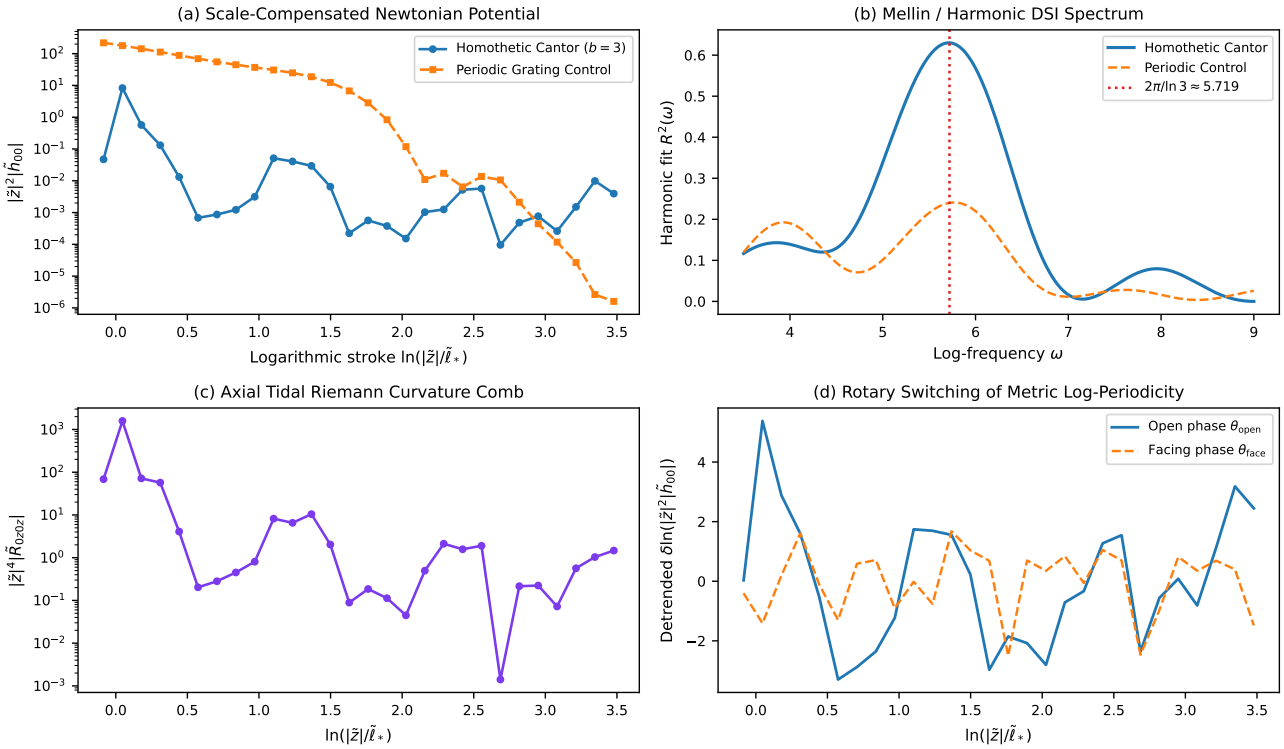


FIG. 5. Imprinting discrete scale invariance onto linearized spacetime geometry via the homothetically staged prefractal CPSC ($b = 3, N = 4$, dimensionless units $\epsilon_P = \ell_P/L$). (a) Numerical BEM–Poisson evaluation of the scale-compensated Newtonian metric modulus $|\tilde{z}|^2|\tilde{h}_{00}(\tilde{\mathbf{x}}(\tilde{z});\tilde{z})|$ swept logarithmically across $\ln(|\tilde{z}|/\tilde{\ell}_*)$ along the collision-free homothetic slot-center ray for the Cantor hierarchy (blue circles) versus the equal-area periodic grating control (orange squares). (b) Harmonic regression spectrum $R^2(\omega)$ of the affine-detrended log-modulus $\delta\ln(|\tilde{z}|^2|\tilde{h}_{00}|)$ for the homothetic Cantor cavity (solid blue, peaking at $\hat{\omega}_1 = 5.7096$) and periodic control (dashed orange) against the exact theoretical DSI frequency $\omega_1 = 2\pi/\ln 3 \approx 5.719$ (vertical dotted red line). (c) Numerical scale-compensated axial tidal Riemann curvature modulus $|\tilde{z}|^4|\tilde{R}_{0z0z}|$ across Cantor stages. (d) Rotary suppression of detrended log-periodic metric oscillations upon rotating from $\theta_{\rm open}$ (solid blue) to $\theta_{\rm face}$ (dashed orange).

obeys [49, 50, 52, 53]

$$\tilde{\mathcal{I}}_{\rm QEI}(\tilde{\tau}) \equiv \int_{-\infty}^{\infty} \tilde{T}_{00}(\tilde{t})\,\frac{2\tilde{\tau}^3/\pi}{(\tilde{t}^2+\tilde{\tau}^2)^2}\,{\rm d}\tilde{t} \geq -\frac{C_{\rm FR}}{\tilde{\tau}^4}, \qquad (31)$$

with $C_{\rm FR} = 3/(32\pi^2)$ per massless scalar ($C_{\rm FR}^{\rm EM} = 3/(16\pi^2)$ for EM; normalized weight $\int \frac{2\tilde{\tau}^3/\pi}{(\tilde{t}^2+\tilde{\tau}^2)^2}{\rm d}\tilde{t} = 1$). Across two decades of $\tilde{\Omega}$, the maximum QEI saturation ratio in our prefractal CPSC is $0.1425 < 1$ (Fig. 6d), confirming strict sub-reflection QEI compliance away from conductor walls.

VIII. PHENOMENON V: CANTOR-STRATIFIED GRAVITOMAGNETIC FRAME DRAGGING FROM BROKEN BOOST INVARIANCE

As clarified in the gravito-electromagnetic (GEM) reviews of Mashhoon [25] and Costa and Natário [26, 27], the term "frame dragging" must never be conflated with viscous fluid dragging or trivial coordinate transformations to a rotating frame. Instead, working strictly in the *non-rotating asymptotic Minkowski laboratory frame* (t, x, y, z), Costa and Natário [27] classify physical frame dragging into three distinct gravitomagnetic levels: (i) the gravitomagnetic 1-form vector potential $\tilde{h}_{0i}$ (governing Sagnac clock-synchronization shifts), (ii) the coordinate-gauge-invariant gravitomagnetic field $\tilde{\mathbf{B}}_g = \tilde{\nabla}\times\tilde{\mathbf{h}}_0$ (governing physical Coriolis-like Lense–Thirring precession $\tilde{\boldsymbol{\Omega}}_{\rm LT} = -\frac{1}{2}\tilde{\mathbf{B}}_g$ of a stationary Fermi–Walker-transported gyroscope relative to distant stars), and (iii) the gravitomagnetic tidal tensor $\tilde{\mathbb{H}}_{ij} = -\frac{1}{2}\partial_{(i}\tilde{B}_{g,j)}$. In linearized Lorenz gauge, $\tilde{h}_{0i}$ is sourced by laboratory momentum density $\tilde{T}^{0i} = -\tilde{T}_{0i}$:

$$\tilde{\nabla}^2\tilde{h}_{0i}(\tilde{\mathbf{x}}) = -16\pi\tilde{T}_{0i}(\tilde{\mathbf{x}}) = +16\pi\tilde{T}^{0i}(\tilde{\mathbf{x}}), \qquad \tilde{\mathbf{B}}_g = \tilde{\nabla}\times\tilde{\mathbf{h}}_0. \qquad (32)$$

Why does a rotating flat-disk Casimir cavity carry zero vacuum angular momentum, whereas a rotating CPSC actively drags stationary gyroscopes? Under a local tangential Lorentz boost $\tilde{v}_\phi = \tilde{\Omega}\tilde{r} \ll 1$ from the boundary rest frame to the inertial laboratory frame, a rest-frame

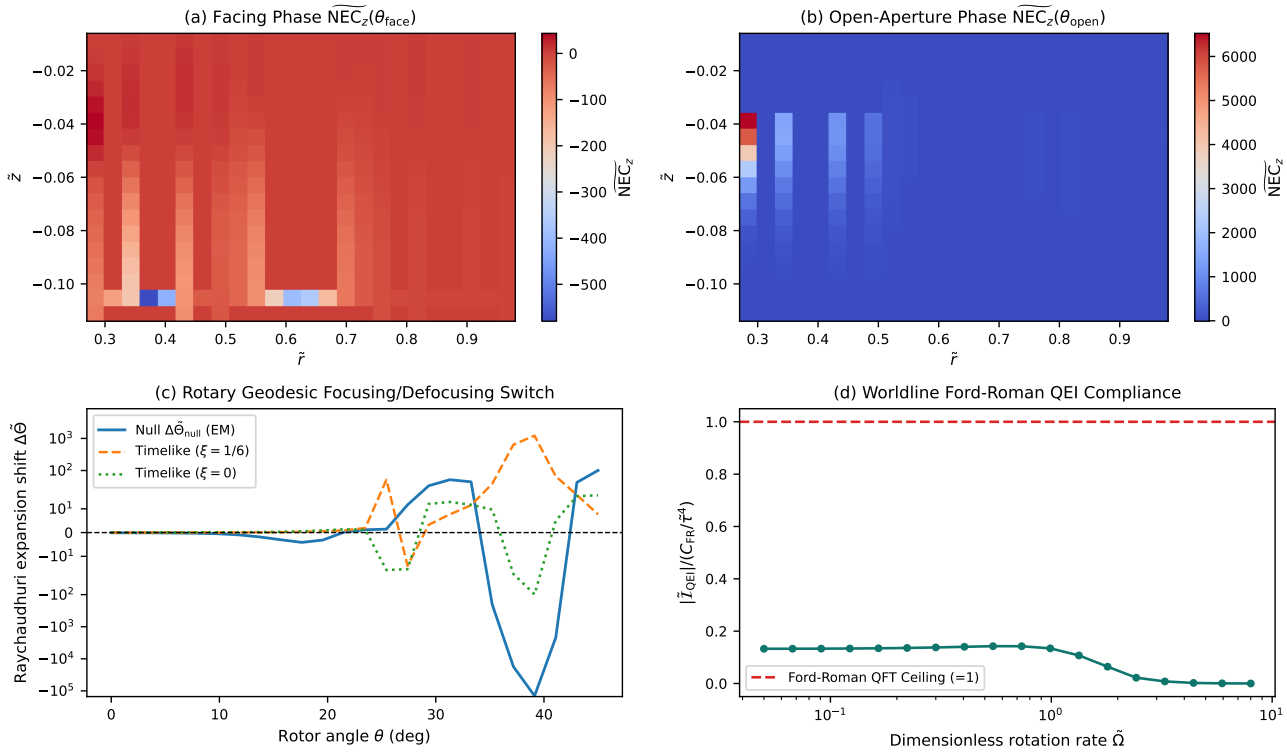


FIG. 6. Rotary switching of general-relativistic energy conditions, Raychaudhuri geodesic focusing, and worldline Quantum Energy Inequality (QEI) compliance in Constellation A $(4, 4)$ ($N = 3$, dimensionless units $\epsilon_P = \ell_P/L$). (a,b) Numerical four-pass BEM voxel maps in the $(\tilde{r}, \tilde{z})$ meridian plane of the axial Null Energy Condition scalar $\widetilde{\mathrm{NEC}}_z = \tilde{T}_{00} + \tilde{T}_{zz}$ in (a) the defocusing facing-sector phase θ_{face} ($\widetilde{\mathrm{NEC}}_z < 0$) and (b) the converging open-aperture phase θ_{open} ($\widetilde{\mathrm{NEC}}_z > 0$). (c) Numerical Raychaudhuri geodesic expansion shifts $\Delta\tilde{\Theta}$ (symmetric logarithmic scale) across $\theta \in [0°, 45°]$ comparing null rays in PEC EM (solid blue) against timelike trajectories for conformal ($\xi = 1/6$, dashed orange) and minimal ($\xi = 0$, dotted green) scalar fields. (d) Numerical Lorentzian-sampled worldline QEI ratio $|\tilde{I}_{\mathrm{QEI}}|/(C_{\mathrm{FR}}/\tilde{\tau}^4)$ versus rotation rate $\tilde{\Omega}$ (solid teal circles) bounded strictly below the analytical Ford–Roman QFT ceiling (= 1, dashed red line).

stress tensor transforms to laboratory momentum density

$$\tilde{T}^{0\phi}_{\mathrm{vac}}(\tilde{\mathbf{x}}) = \frac{\tilde{v}_\phi}{1-\tilde{v}_\phi^2}\big(\tilde{\rho}_{\mathrm{vac}} + \tilde{P}_\phi\big) \simeq \tilde{\Omega}\tilde{r}\big(\tilde{T}_{00} + \tilde{T}_{\phi\phi}\big) + \mathcal{O}(\tilde{\Omega}^3), \tag{33}$$

corresponding to net axial vacuum angular momentum $\tilde{J}^{\mathrm{vac}}_z = \tilde{\Omega}\int \tilde{r}^2(\tilde{T}_{00}+\tilde{T}_{\phi\phi})\mathrm{d}^3\tilde{x}$. For planar plates (2), boost invariance enforces $\tilde{T}_{00} + \tilde{T}_{\phi\phi} = \tilde{\rho}_{\mathrm{vac}} + \tilde{P}_{\parallel} = 0$ identically (numerical enthalpy residual 0), so $\tilde{T}^{0\phi}_{\mathrm{vac}} = 0$ and $\tilde{J}^{\mathrm{vac}}_z = 0$. In Constellation C, radial and chiral sector walls break ϕ-boost symmetry, producing a sign-alternating azimuthal vacuum enthalpy $\tilde{w}_\phi \equiv \tilde{T}_{00} + \tilde{T}_{\phi\phi} \neq 0$ across the 2^{m-1} Cantor slot tracks. Solving Eq. (32) in the inertial laboratory frame—thereby solving the active counterpart to Sorge's and Zhang's passive stationary Kerr/rotating-background test-field problems [46, 47]—yields a Cantor-stratified Lense–Thirring vortex lattice with 4 radial vorticity sign reversals, peak normalized gyroscope precession $|\tilde{\Omega}^{\mathrm{vac}}_{\mathrm{LT},z}|/\tilde{\Omega} = 22931.2037$ at $N = 3$, and a 23.0732× enhancement in root-mean-square gravitomagnetic vorticity over the matched periodic grating (Fig. 7).

IX. DISCUSSION: COVARIANT CONSISTENCY, EQUIVALENCE PRINCIPLE, AND PLANCK-SCALE MAGNITUDE

A. Role of Constraint Stresses and Elimination of Coordinate Artifacts

Evaluating linearized Einstein or Komar integrals over $\langle\hat{T}_{\mu\nu}\rangle_{\mathrm{vac}}$ without mechanical support stresses $\tilde{T}^{\mathrm{con}}_{\mu\nu}$ violates $\partial_\mu\tilde{T}^{\mu\nu}_{\mathrm{tot}} = 0$ and introduces gauge artifacts [29, 32, 38]. Enforcing Eq. (15) explicitly across all five phenomena clarifies how boundary reactions enter each sector:

1. In the $(6, 4)$ Vernier radiator (Sec. IV), elastic support stresses balance local pressures; however, because $\tilde{\mathcal{Q}}^{\mathrm{tot}}_{ij}$ integrates energy density $\tilde{T}_{00}$—where rigid C_6 and C_4 mass distributions have identically zero $m = 2$ quadrupole and elastic strain energy scales as $\tilde{P}^2/Y_{\mathrm{rigid}} \to 0$ in the rigid limit—the 6Ω spin-2 wave is genuinely sourced by the sweeping $m = 2$ Casimir vacuum energy wave, verified to virial error 1.45×10^{-15}.

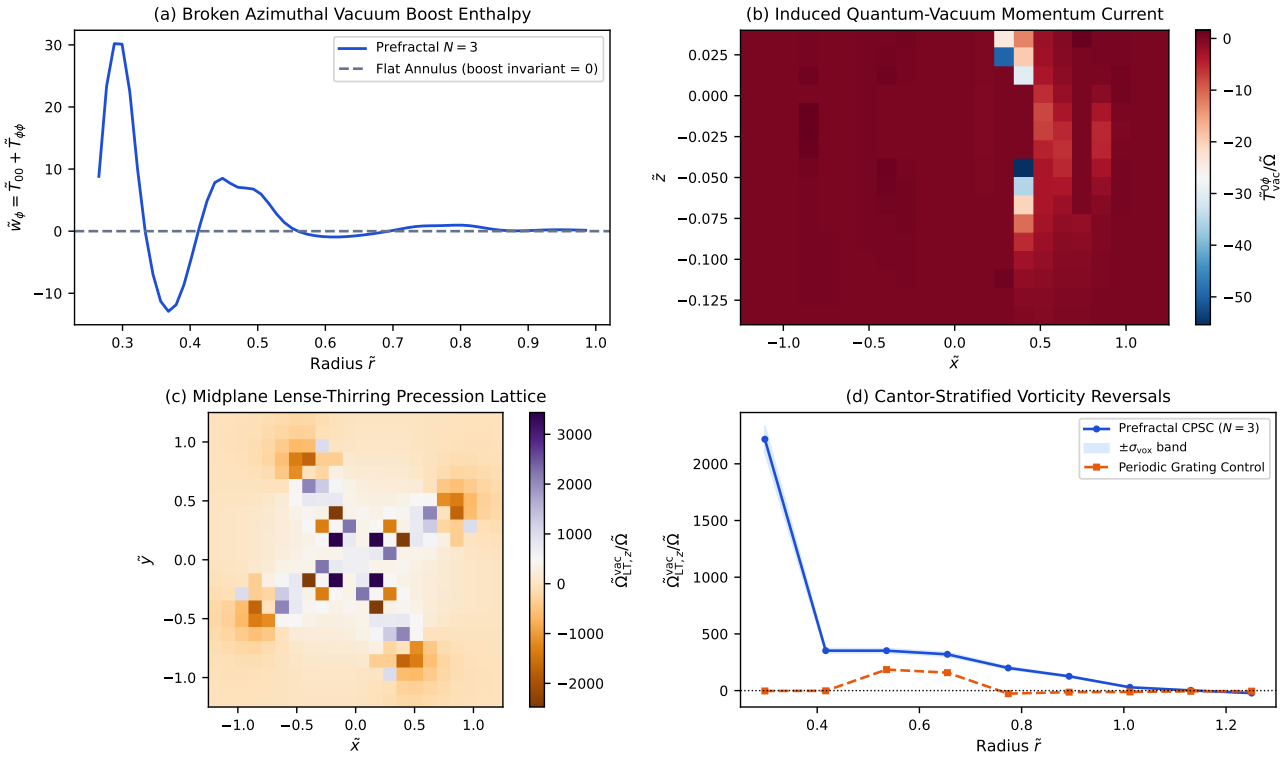


FIG. 7. Gauge-invariant gravitomagnetic Lense–Thirring precession generated by broken azimuthal Lorentz boost invariance in Constellation C (chiral $(4,4)$, $N = 3$, dimensionless units $\epsilon_P = \ell_P/L$). (a) Numerical radial profile of the rest-frame azimuthal vacuum enthalpy $\tilde{w}_\phi = \tilde{T}_{00} + \tilde{T}_{\phi\phi}$ for the prefractal CPSC (solid blue) versus the boost-invariant flat-annulus control ($\tilde{w}_\phi = 0$, dashed gray). (b) Cross-sectional $(\tilde{x}, \tilde{z})$ map of the laboratory-frame azimuthal vacuum momentum current $\tilde{T}^{0\phi}_{\rm vac}/\tilde{\Omega}$. (c) Midplane $(\tilde{x}, \tilde{y})$ lattice of the dimensionless Lense–Thirring gyroscope precession rate $\tilde{\Omega}^{\rm vac}_{{\rm LT},z}/\tilde{\Omega} = -\frac{1}{2}\tilde{B}^{\rm vac}_{g,z}/\tilde{\Omega}$ obtained from the 3D spectral GEM Poisson solver (32). (d) Radial slice of $\tilde{\Omega}^{\rm vac}_{{\rm LT},z}/\tilde{\Omega}$ with $\pm\sigma_{\rm vox}$ discretization band (shaded blue) showing 4 Cantor vorticity sign reversals compared against the matched periodic grating control (dashed orange squares).

TABLE I. Summary of the five active semiclassical gravitational phenomena of the Canonical Prefractal Sector Cavity (CPSC) in Planck-normalized dimensionless units ($\epsilon_P = \ell_P/L$), contrasting analytical/numerical flat-annulus limits, equal-area single-scale periodic controls, and prefractal Cantor hierarchies ($b = 3$).

Gravitational Phenomenon	Governing Coupling	Flat Annulus	Periodic Control	Prefractal CPSC
I(a). Spin-2 Vernier GWs $(6,4)$	$\ddot{\tilde{\mathcal{Q}}}^{\rm tot}_{ij}$ ($\tilde{h}_+$ amp.)	0 (axisym.)	0.2969	0.1145 (0.3857×)
I(b). Spin-0 Scalaron Wave $(4,4)$	$\tilde{\mathcal{T}}_{\rm eff}(t)$ $(R + \alpha R^2, \alpha > 0)$	0	Single-band	27.122 (Yukawa 9.80×10^{-4}×)
II. Force-Free Curvature ($\tilde{F}_z = 0$)	$\tilde{R}_{\rm eff}(\theta_0)$, $\theta_0 = 25.3298°$	Forbidden ($\tilde{F}_z < 0$)	Single crossover	−2317.8935 (lock 3.0018)
III. Log-Periodic Metric Comb	$\hat{\omega}_1$ (R^2 at $2\pi/\ln b$)	Power-law (0)	$R^2 = 0.2411$	$\hat{\omega}_1 = 5.7096$ ($R^2 = 0.6301$)
IV. NEC/SEC & Geodesic Switch	$\widetilde{\rm NEC}_z$ / QEI sat.	Violated ($\forall\tilde{\mathbf{x}}$)	Single switch	$[-579.895, 6523.7421]$; QEI 0.1425
V. Vacuum Frame Dragging	$\lvert\tilde{\Omega}^{\rm vac}_{{\rm LT},z}\rvert/\tilde{\Omega}$	0 (boost inv.)	Ref. vortex	22931.2037 (23.0732×, 4 rev.)

2. In the stationary Tolman–Komar sector (Sec. V), exterior support stresses enforce $\int_{\mathbb{R}^3} \tilde{P}_{i,\rm tot} {\rm d}^3\tilde{x} = 0$ globally so $\Delta\tilde{M}_{\rm Komar}(\infty) = \tilde{E}_C(\theta)$ [32, 38], while leaving interior coarse-grained curvature $\tilde{R}_{\rm eff}(\theta_0)$ at zero axial force unscreened inside the open gap.

B. Physical Scale Estimates and Analog-Gravity Connections

Restoring physical constants at an illustrative outer radius $L = 1$ μm ($\epsilon_P^2 = \ell_P^2/L^2 = 2.612 \times 10^{-58}$) underscores the purely foundational nature of these effects in real spacetime: static metric perturbations scale as $|h_{\mu\nu}| = \epsilon_P^2|\tilde{h}_{\mu\nu}| \sim 10^{-56}$–$10^{-54}$, coarse-grained Ricci curvatures scale as $|R_{\rm eff}| = (\epsilon_P^2/L^2)|\tilde{R}_{\rm eff}| \sim 10^{-43}$ m^{-2}, and

Vernier strains scale as $h_+ \sim \epsilon_P^2 (L/r_{\rm obs})(\Omega L/c)^2 |\tilde{Q}_2|$, far below direct gravitational-wave detection thresholds even for synchronized $N_{\rm arr} \times N_{\rm arr}$ arrays. Nevertheless, in acoustic Bose–Einstein condensate and optical metamaterial analog-gravity platforms where effective propagation speeds $c_{\rm eff} \ll c$ and effective couplings replace ℓ_P, analogous boundary-induced effective-trace and boost-breaking vorticity phenomena offer promising targets for laboratory simulation.

X. CONCLUSION

We have formulated a self-contained semiclassical field-theoretic and numerical framework for prefractal Casimir gravitation at zero temperature, uniting three-dimensional boundary-element scattering with linearized Einstein, Starobinsky $f(R)$, and gravito-electromagnetic theory. Strictly separating the cavity-integrated macroscopic effective trace [13] from local curved-spacetime conformal trace anomalies [8, 9] and enforcing covariant stress-energy conservation [32, 38], we demonstrated five active gravitational backreaction phenomena in the Canonical Prefractal Sector Cavity (Table I): (1) symmetry-filtered 6Ω Vernier spin-2 gravitational waves and Yukawa-screened $l = 0$ Starobinsky scalaron waves; (2) an axial-force-free macroscopic curvature crossover respecting global Tolman–Komar WEP mass recovery; (3) a switchable discrete-scale-invariant log-periodic Newtonian potential and tidal Riemann comb; (4) rotary NEC/SEC geodesic focusing/defocusing switching strictly bounded by Ford–Roman and Fewster worldline QEIs; and (5) Cantor-stratified Lense–Thirring frame dragging driven by broken azimuthal boost invariance.

Several caveats and future directions merit emphasis. First, our calculations assume idealized zero-temperature ($T = 0$ K) lossless Dirichlet and PEC boundaries in the adiabatic slow-rotation regime ($\Omega L/c \ll 1$); incorporating finite-conductivity Lifshitz dielectric dispersion, nonzero temperature ($T > 0$ K, where the fractal thermal trace $\rho_{\rm th}(3/d_s - 1)$ enters [13, 55]), and non-adiabatic dynamical Casimir particle creation at $\Omega L/c \sim \mathcal{O}(1)$ [45] represents a natural theoretical extension. Second, while real-spacetime amplitudes are suppressed by $\epsilon_P^2 \sim 10^{-58}$ at micrometer length scales, mapping the Vernier selection rules and broken-boost enthalpy currents onto metamaterial and superfluid analog-gravity systems provides a concrete avenue for future experimental exploration.

DATA AVAILABILITY

All simulation scripts, three-dimensional boundary-element and spectral linearized gravity solvers (implemented in Python 3.14.0 using NumPy 2.4.6 [70] and SciPy 1.16.3 [71]), provenance ledgers, figure sidecars, and raw JSON datasets supporting the findings of this study are included in the standalone `fractal_casimir_gravity` software package (https://github.com/gogordon-iu/FractalCasimir3D, branch `fractal-casimir-gravity`) and are also available from the author upon reasonable request.

ACKNOWLEDGMENTS

Supercomputing resource support was provided by the BigRed 200 facilities at Indiana University Bloomington. In accordance with American Physical Society policy on artificial intelligence tools, the author discloses that the Google DeepMind Antigravity agentic coding and research environment (powered by Gemini 3.1 Pro) was used to assist with Python solver implementation, automated unit/regression testing, figure generation scripts, and LaTeX manuscript editing. All analytical derivations and tensor identities were independently verified by the author and cross-checked symbolically in SymPy, and all numerical outputs were validated against exact parallel-plate Casimir limits, relativistic virial conservation laws, systematic mesh/quadrature convergence benchmarks, and automated provenance audits.

Appendix A: Covariant Virial Identity and Conservation of the Total Quadrupole Tensor

In linearized general relativity around Minkowski spacetime, Lorenz gauge $\partial^\mu \bar{h}_{\mu\nu} = 0$ requires $\partial_\mu T^{\mu\nu}_{\rm tot} = 0$:

$$\frac{\partial \tilde{T}^{00}_{\rm tot}}{\partial \tilde{t}} + \partial_j \tilde{T}^{j0}_{\rm tot} = 0, \qquad \frac{\partial \tilde{T}^{0i}_{\rm tot}}{\partial \tilde{t}} + \partial_j \tilde{T}^{ji}_{\rm tot} = 0. \tag{A1}$$

Multiplying the energy continuity equation by $\tilde{x}^i \tilde{x}^j$, integrating over a ball $\mathcal{B}_R$ enclosing the isolated cavity-plus-support assembly ($\tilde{T}^{\mu\nu}_{\rm tot} = 0$ on $\partial\mathcal{B}_R$), and integrating by parts twice using momentum continuity yields the relativistic second-moment virial theorem:

$$\frac{\mathrm{d}^2}{\mathrm{d}\tilde{t}^2} \int_{\mathcal{B}_R} \tilde{T}^{00}_{\rm tot}(\tilde{\mathbf{x}}, \tilde{t})\, \tilde{x}^i \tilde{x}^j \, \mathrm{d}^3\tilde{x} = 2 \int_{\mathcal{B}_R} \tilde{T}^{ij}_{\rm tot}(\tilde{\mathbf{x}}, \tilde{t})\, \mathrm{d}^3\tilde{x}. \tag{A2}$$

In the ideal rigid-support limit ($Y_{\rm rigid} \to \infty$), elastic displacements scale as $\delta\tilde{u} \sim \tilde{P}/Y_{\rm rigid} \to 0$, so elastic strain energy vanishes ($\tilde{T}^{00}_{\rm con,elas} \sim \tilde{P}^2/Y_{\rm rigid} \to 0$) while constraint stresses $\tilde{T}^{ij}_{\rm con} \sim \mathcal{O}(\tilde{P})$ satisfy $\partial_j \tilde{T}^{ji}_{\rm con} = -\partial_j \tilde{T}^{ji}_{\rm vac} - \partial_{\tilde{t}} \tilde{T}^{0i}_{\rm tot}$ at boundary walls. For static configurations ($\partial_{\tilde{t}} = 0$), Eq. (A2) gives $\int \tilde{T}^{ij}_{\rm con} \mathrm{d}^3\tilde{x} = -\int \tilde{T}^{ij}_{\rm vac} \mathrm{d}^3\tilde{x}$ (Laue's theorem), ensuring $\Delta\tilde{M}_{\rm Komar}(\infty) = \tilde{E}_C$ [32, 38]. During steady rotation $\theta(\tilde{t}) = \tilde{\Omega}\tilde{t}$ of the $(6, 4)$ Vernier cavity, $\tilde{T}^{00}_{\rm vac}$ carries the $m = 2$ wave (21) whereas rigid C_6 and C_4 mass densities carry zero $m = 2$ harmonic; by Eq. (A2), $\int (\tilde{T}^{ij}_{\rm vac} + \tilde{T}^{ij}_{\rm con})^{\rm TF} \mathrm{d}^3\tilde{x} = \frac{1}{2} \ddot{\tilde{\mathcal{Q}}}^{\rm vac}_{ij}$, verified numerically to relative error 1.45×10^{-15}.

Appendix B: Scattering-Operator Derivation of the Vernier Multipole Selection Rule and GW Reciprocity

Let $\mathcal{U}(\alpha)$ denote the spatial rotation operator about the z axis by angle α. For an upper boundary $\partial\Omega_{\rm U}(\theta)$ with discrete $C_{M_{\rm up}}$ symmetry rotated by θ and a stationary lower boundary $\partial\Omega_{\rm L}$ with $C_{M_{\rm low}}$ symmetry, the single-body Euclidean scattering matrices satisfy $\mathcal{U}(2\pi/M_{\rm low})\mathbf{\Gamma}_{\rm LL}\mathcal{U}^{-1}(2\pi/M_{\rm low}) = \mathbf{\Gamma}_{\rm LL}$ and $\mathbf{\Gamma}_{\rm UU}(\theta) = \mathcal{U}(\theta)\mathbf{\Gamma}_{\rm UU}(0)\mathcal{U}^{-1}(\theta)$ with $\mathcal{U}(2\pi/M_{\rm up})\mathbf{\Gamma}_{\rm UU}(0)\mathcal{U}^{-1}(2\pi/M_{\rm up}) = \mathbf{\Gamma}_{\rm UU}(0)$. Expanding the off-diagonal propagators in cylindrical azimuthal harmonics $e^{i\ell\phi}$ ($\ell \in \mathbb{Z}$), $\mathbf{\Gamma}^{-1}_{\rm UU}(\theta)$ couples $\ell \to \ell + pM_{\rm up}$ with phase $e^{-ipM_{\rm up}\theta}$ ($p \in \mathbb{Z}$), whereas $\mathbf{\Gamma}^{-1}_{\rm LL}$ couples $\ell' \to \ell' - qM_{\rm low}$ ($q \in \mathbb{Z}$). Inserting this Neumann/Born expansion of $\mathbf{\Gamma}^{-1}(\tilde{\kappa};\theta)$ into the coincidence Green's correlator (18) shows that every term in the renormalized vacuum energy density takes the exact double-Bloch series form

$$\Delta\tilde{T}^{\rm vac}_{00}(\tilde{r},\phi,\tilde{z};\theta) = \sum_{p=-\infty}^{\infty}\sum_{q=-\infty}^{\infty} \mathcal{A}_{p,q}(\tilde{r},\tilde{z}) \times e^{i(pM_{\rm up}-qM_{\rm low})\phi - ipM_{\rm up}\theta}. \quad \text{(B1)}$$

Projecting onto radiative spin-s spherical/cylindrical multipoles $e^{\mp is\phi}$ ($s = 2$ for spin-2 quadrupole radiation) enforces the exact selection rule

$$pM_{\rm up} - qM_{\rm low} = \pm s, \qquad \omega_{\rm rad} = |p|M_{\rm up}\Omega. \quad \text{(B2)}$$

For $(M_{\rm up}, M_{\rm low}) = (6,4)$ and $s = 2$, the minimal non-zero integers $(p,q) = (\pm1,\pm1)$ satisfy $6(\pm1) - 4(\pm1) = \pm2$ and radiate at $\omega_{\rm GW} = 6\Omega$, whereas any non-interacting rigid body ($q = 0$ or $p = 0$) requires $6p = \pm2$ or $-4q = \pm2$, which has no integer solution. Conversely, if the $(6,4)$ cavity rotates inside an incoming circularly polarized plane gravitational wave $h^{\rm ext}_{+}(t) = h_0\cos(\omega_{\rm GW}t)$, the linear interaction Hamiltonian $H_{\rm int}(t) = -\frac{1}{2}\int h^{\rm ext}_{ij}(t)T^{ij}_{\rm tot}(\mathbf{x},t)\mathrm{d}^3x = -\frac{1}{4}\ddot{h}^{\rm ext}_{ij}(t)\mathcal{Q}^{\rm vac}_{ij}(t)$ yields a secular phase-locked energy transfer (and permanent memory shift in the sense of Sorge [44]) selectively when $\omega_{\rm GW} = 6\Omega$, while reducing in the high-frequency non-adiabatic limit ($\omega_{\rm GW} \to 2\omega_{\rm cav}$) to the parametric dynamical Casimir resonance derived by de Oliveira *et al.* [45].

[1] N. D. Birrell and P. C. W. Davies, *Quantum Fields in Curved Space* (Cambridge University Press, Cambridge, 1982).
[2] R. M. Wald, *General Relativity* (University of Chicago Press, Chicago, 1984).
[3] M. Cerdonio and C. Rovelli, Casimir effects are not an experimental demonstration that free vacuum gravitates: Connections to the cosmological constant problem, Int. J. Mod. Phys. D **24**, 1544020 (2015).
[4] C. W. Misner, K. S. Thorne, and J. A. Wheeler, *Gravitation* (W. H. Freeman, San Francisco, 1973).
[5] H. B. G. Casimir, On the attraction between two perfectly conducting plates, Proc. K. Ned. Akad. Wet. **51**, 793 (1948).
[6] M. Bordag, G. L. Klimchitskaya, U. Mohideen, and V. M. Mostepanenko, *Advances in the Casimir Effect* (Oxford University Press, Oxford, 2009).
[7] L. S. Brown and G. J. Maclay, Vacuum stress between conducting plates: An image solution, Phys. Rev. **184**, 1272 (1969).
[8] M. J. Duff, Twenty years of the Weyl anomaly, Class. Quantum Grav. **11**, 1387 (1994).
[9] M. R. Setare and A. H. Rezaeian, Trace anomaly and Casimir effect, Mod. Phys. Lett. A **15**, 2159 (2000).
[10] C. G. Callan, Jr., S. Coleman, and R. Jackiw, A new improved energy-momentum tensor, Ann. Phys. (N.Y.) **59**, 42 (1970).
[11] D. Deutsch and P. Candelas, Boundary effects in quantum field theory, Phys. Rev. D **20**, 3063 (1979).
[12] K. A. Milton, S. A. Fulling, P. Parashar, P. Kalauni, and T. Murphy, Stress tensor for a scalar field in a spatially varying background potential: Divergences, renormalization, anomalies, and Casimir forces, Phys. Rev. D **93**, 085017 (2016).
[13] G. Gordon, Effective trace framework for self-similar Casimir systems, Phys. Rev. D **113**, 125013 (2026).
[14] L. Buoninfante, G. Lambiase, L. Petruzziello, and A. Stabile, Casimir effect in quadratic theories of gravity, Eur. Phys. J. C **79**, 41 (2019).
[15] A. A. Starobinsky, A new type of isotropic cosmological models without singularity, Phys. Lett. B **91**, 99 (1980).
[16] T. Damour and G. Esposito-Farèse, Tensor-multi-scalar theories of gravitation, Class. Quantum Grav. **9**, 2093 (1992).
[17] A. De Felice and S. Tsujikawa, $f(R)$ theories, Living Rev. Relativ. **13**, 3 (2010).
[18] C. P. L. Berry and J. R. Gair, Linearized $f(R)$ gravity: Gravitational radiation and Solar System tests, Phys. Rev. D **83**, 104022 (2011).
[19] B. Pourhassan, Rotating Casimir wormholes in $f(R)$ gravity: A modified gravity extension of exotic spacetime models, Eur. Phys. J. C **85**, 977 (2025).
[20] A. K. Raychaudhuri, Relativistic cosmology. I, Phys. Rev. **98**, 1123 (1955).
[21] E.-A. Kontou and K. Sanders, Energy conditions in general relativity and quantum field theory, Class. Quantum Grav. **37**, 193001 (2020).

[22] A. Manjavacas and F. J. García de Abajo, Vacuum friction in rotating particles, Phys. Rev. Lett. **105**, 113601 (2010).

[23] M. F. Maghrebi, R. L. Jaffe, and M. Kardar, Spontaneous emission by rotating objects: A scattering approach, Phys. Rev. Lett. **108**, 230403 (2012).

[24] I. Ciufolini and E. C. Pavlis, A confirmation of the general relativistic prediction of the Lense–Thirring effect, Nature **431**, 958 (2004).

[25] B. Mashhoon, Gravitoelectromagnetism: A brief review, in *The Measurement of Gravitomagnetism: A Challenging Enterprise*, edited by L. Iorio (Nova Science Publishers, New York, 2007) pp. 29–39.

[26] L. F. O. Costa and J. Natário, Gravito-electromagnetic analogies, Gen. Relativ. Gravit. **46**, 1792 (2014).

[27] L. F. O. Costa and J. Natário, Frame-dragging: Meaning, myths, and misconceptions, Universe **7**, 388 (2021).

[28] F. Sorge, Casimir effect in a weak gravitational field, Class. Quantum Grav. **22**, 5109 (2005).

[29] G. Bimonte, E. Calloni, G. Esposito, and L. Rosa, Energy-momentum tensor for a Casimir apparatus in a weak gravitational field, Phys. Rev. D **74**, 085011 (2006).

[30] G. Bimonte, E. Calloni, G. Esposito, and L. Rosa, Relativistic mechanics of Casimir apparatuses in a weak gravitational field, Phys. Rev. D **76**, 025008 (2007).

[31] A. P. C. M. Lima, G. Alencar, and R. R. Landim, Null second order corrections to Casimir energy in weak gravitational field: The Schwinger's approach, J. Cosmol. Astropart. Phys. **2021** (01), 056.

[32] S. A. Fulling, K. A. Milton, P. Parashar, A. Romeo, K. V. Shajesh, and J. Wagner, How does Casimir energy fall?, Phys. Rev. D **76**, 025004 (2007).

[33] K. A. Milton, K. V. Shajesh, S. A. Fulling, and P. Parashar, How does Casimir energy fall? IV. Gravitational interaction of regularized quantum vacuum energy, Phys. Rev. D **89**, 064027 (2014).

[34] J. D. Bekenstein, If vacuum energy can be negative, why is mass always positive?: Uses of the subdominant trace energy condition, Phys. Rev. D **88**, 125005 (2013).

[35] M. Cerdonio and C. Rovelli, Casimir cavities do not fly (2015), arXiv:1406.1105 [gr-qc].

[36] E. Calloni, M. De Laurentis, R. De Rosa, F. Garufi, L. Rosa, L. Di Fiore, G. Esposito, C. Rovelli, P. Ruggi, and F. Tafuri, Towards weighing the condensation energy to ascertain the Archimedes force of vacuum, Phys. Rev. D **90**, 022002 (2014).

[37] E. Calloni, S. Caprara, M. De Laurentis, G. Esposito, M. Grilli, E. Majorana, G. P. Pepe, S. Petrarca, P. Puppo, P. Rapagnani, *et al.*, The Archimedes experiment, Nucl. Instrum. Methods Phys. Res., Sect. A **824**, 646 (2016).

[38] F. Sorge, Quasi-local Casimir energy and vacuum buoyancy in a weak gravitational field, Class. Quantum Grav. **38**, 025009 (2021).

[39] F. Sorge and J. H. Wilson, Casimir effect in free fall towards a Schwarzschild black hole, Phys. Rev. D **100**, 105007 (2019).

[40] J. H. Wilson, F. Sorge, and S. A. Fulling, Tidal and nonequilibrium Casimir effects in free fall, Phys. Rev. D **101**, 065007 (2020).

[41] M. N. Chernodub, Rotating Casimir systems: Magnetic-field-enhanced perpetual motion, possible realization in doped nanotubes, and laws of thermodynamics, Phys. Rev. D **87**, 025021 (2013).

[42] M. Schaden, Quantization and renormalization and the Casimir energy of a scalar field interacting with a rotating ring (2012), arXiv:1211.2740 [quant-ph].

[43] A. Flachi and M. Edmonds, Quantum vacuum, rotation, and nonlinear fields, Phys. Rev. D **107**, 025008 (2023).

[44] F. Sorge, Gravitational memory of Casimir effect, Phys. Rev. D **108**, 104003 (2023).

[45] G. de Oliveira, T. H. Moreira, and L. C. Céleri, Dynamical Casimir effect under the action of gravitational waves, Entropy **28**, 177 (2026).

[46] F. Sorge, Casimir energy in Kerr space-time, Phys. Rev. D **90**, 084050 (2014).

[47] A. Zhang, Theoretical analysis of Casimir and thermal Casimir effect in stationary space-time, Phys. Lett. B **773**, 125 (2017).

[48] R. Garattini and A. G. Tzikas, Rotating Casimir wormholes, Eur. Phys. J. C **85**, 336 (2025).

[49] L. H. Ford and T. A. Roman, Averaged energy conditions and quantum inequalities, Phys. Rev. D **51**, 4277 (1995).

[50] L. H. Ford and T. A. Roman, Restrictions on negative energy density in flat spacetime, Phys. Rev. D **55**, 2082 (1997).

[51] C. J. Fewster and S. P. Eveson, Bounds on negative energy densities in flat spacetime, Phys. Rev. D **58**, 084010 (1998).

[52] C. J. Fewster and M. J. Pfenning, Quantum energy inequalities and local covariance I: Globally hyperbolic spacetimes, J. Math. Phys. **47**, 082303 (2006).

[53] C. J. Fewster, Quantum energy inequalities, in *Wormholes, Warp Drives and Energy Conditions*, Fundamental Theories of Physics, Vol. 189, edited by F. S. N. Lobo (Springer, Cham, 2017) pp. 215–254.

[54] N. Graham and K. D. Olum, Negative energy densities in quantum field theory with a background potential, Phys. Rev. D **67**, 085014 (2003).

[55] E. Akkermans, G. V. Dunne, and A. Teplyaev, Thermodynamics of photons on fractals, Phys. Rev. Lett. **105**, 230407 (2010).

[56] G. Calcagni, Fractal universe and quantum gravity, Phys. Rev. Lett. **104**, 251301 (2010).

[57] H. Cheng, The Casimir effect for parallel plates in the spacetime with a fractal extra compactified dimension, Int. J. Theor. Phys. **52**, 3229 (2013).

[58] M. L. Lapidus, Fractal drum, inverse spectral problems for elliptic operators and a partial resolution of the Weyl–Berry conjecture, Trans. Am. Math. Soc. **325**, 465 (1991).

[59] D. Sornette, Discrete-scale invariance and complex dimensions, Phys. Rep. **297**, 239 (1998).

[60] M. W. Choptuik, Universality and scaling in gravitational collapse of a massless scalar field, Phys. Rev. Lett. **70**, 9 (1993).

[61] K. V. Shajesh, I. Brevik, I. Cavero-Peláez, and P. Parashar, Casimir energies of self-similar plate configurations, Phys. Rev. D **94**, 065003 (2016).

[62] K. V. Shajesh, P. Parashar, I. Cavero-Peláez, J. Kocik, and I. Brevik, Casimir energy of Sierpinski triangles, Phys. Rev. D **96**, 105010 (2017).

[63] O. Kenneth and I. Klich, Opposites attract: A theorem about the Casimir force, Phys. Rev. Lett. **97**, 160401 (2006).

[64] M. Levin, A. P. McCauley, A. W. Rodriguez, M. T. H. Reid, and S. G. Johnson, Casimir repulsion between metallic objects in vacuum, Phys. Rev. Lett. **105**, 090403

(2010).
[65] M. T. H. Reid, A. W. Rodriguez, J. White, and S. G. Johnson, Efficient computation of Casimir interactions between arbitrary 3D objects, Phys. Rev. Lett. **103**, 040401 (2009).
[66] M. T. H. Reid, J. White, and S. G. Johnson, Fluctuating surface currents: An algorithm for efficient prediction of Casimir interactions among arbitrary materials in arbitrary geometries, Phys. Rev. A **88**, 022514 (2013).
[67] A. W. Rodriguez, A. P. McCauley, J. D. Joannopoulos, and S. G. Johnson, Casimir forces in the time domain: Theory, Phys. Rev. A **80**, 012115 (2009).
[68] K. S. Thorne, Multipole expansions of gravitational radiation, Rev. Mod. Phys. **52**, 299 (1980).
[69] D. V. Vassilevich, Heat kernel expansion: User's manual, Phys. Rep. **388**, 279 (2003).
[70] C. R. Harris, K. J. Millman, S. J. van der Walt, R. Gommers, P. Virtanen, D. Cournapeau, E. Wieser, J. Taylor, S. Berg, N. J. Smith, *et al.*, Array programming with NumPy, Nature **585**, 357 (2020).
[71] P. Virtanen, R. Gommers, T. E. Oliphant, M. Haberland, T. Reddy, D. Cournapeau, E. Burovski, P. Peterson, W. Weckesser, J. Bright, *et al.*, SciPy 1.0: Fundamental algorithms for scientific computing in Python, Nat. Methods **17**, 261 (2020).
[72] R. C. Tolman, On the use of the energy-momentum principle in general relativity, Phys. Rev. **35**, 875 (1930).
[73] A. Komar, Covariant conservation laws in general relativity, Phys. Rev. **113**, 934 (1959).